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The Independent and Stage-Specific Role of Skeletal Muscle in Glycaemic Control

A Cross-Sectional Analysis of the NHANES 2011-2018 Data

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Chapter 1

Abstract

The global prevalence of diabetes has increased substantially in recent decades. Effectively addressing this trend requires a better understanding of the factors associated with the disease. This study examined the independent and potentially stage-specific associations between skeletal muscle mass and glycaemic control in a large, nationally representative US sample.

The analytical sample consisted of 12,178 adults between the ages of 18 and 59 from the 2011-18 cycles of NHANES. Blood glucose was measured using glycated haemoglobin (HbA1c), and skeletal muscle mass was quantified using the appendicular lean mass index (ALMI). After handling missing data with multiple imputation, survey-weighted logistic regression was employed in a dual-model framework by modelling the “Onset” of elevated HbA1c ($\geq 5.7\%$) and “Progression” to clinical diabetes ($\geq 6.5\%$) individually.

Stage-specific associations were found between skeletal muscle mass and glycaemic control after comprehensive adjustment for confounding. Increases in ALMI were not independently associated with the onset of dysglycaemia but were linked to lower odds of the progression to diabetes, with a $1\text{kg}/\text{m}^2$ increase corresponding to a 21.2% reduction in odds (OR = 0.79, 95% CI [0.68, 0.91]). The association with onset became evident only after controlling for central adiposity, whereas the association with progression emerged after accounting for both central and overall adiposity.

In conclusion, the independent protective association of skeletal muscle with diabetes risk appears to be specific to the later stages of glycaemic dysregulation. These findings highlight the importance of comprehensive adjustment for both central and overall adiposity and suggest that the relationship between skeletal muscle mass and glycaemia may vary across different stages of dysregulation.

Chapter 2

Introduction

Epidemiological data indicate a rising prevalence of both type 1 and type 2 diabetes mellitus in recent years. The World Health Organisation states that the number of individuals affected by diabetes rose by 315% from 1990 to 2022 [47]. This is a significant public health concern as diabetes is recognised as a leading cause of morbidity and mortality on a global scale [14]. Prediabetes, an intermediate metabolic state characterised by elevated blood glucose levels below the diagnostic threshold for type 2 diabetes but associated with a substantially increased risk of progression, is also rising in global prevalence. Projections indicate that by 2030, more than 470 million individuals will be classified as prediabetic [52]. Of these, an estimated 5–10% are expected to progress to type 2 diabetes annually, suggesting that between 23.5 and 47 million or more individuals may develop diabetes by 2030. Understanding the key, modifiable risk factors for these conditions is essential for developing effective prevention strategies.

One of the most commonly known risk factors for diabetes is obesity [34]. Obesity is characterised by excessive body fat and is frequently identified using measurements such as Body Mass Index (BMI). However, simple metrics such as BMI are often insufficient as they do not distinguish between fat and muscle mass and do not describe the distribution of fat in the body. This is a concern as a more accurate understanding of metabolic risk requires a focus on the interplay between different aspects of body composition such as adiposity and muscle. The distribution of fat is also important, with studies showing that visceral fat, in particular, has a distinct association with diabetes risk compared with overall adiposity [34].

A specific type of muscle mass has been shown to be protective against the prevalence of diabetes - skeletal muscle [51]. Skeletal muscle is not simply a tissue used for movement, but is also the body's largest metabolic organ [48]. Physiologically, it is the primary site for postprandial glucose disposal, acting as a metabolic sink and accounting for approximately over 80% of post-meal glucose uptake from the blood [42]. Consequently, insulin sensitivity within skeletal muscle is a key determinant of whole-body glucose homeostasis. When this process is impaired, it becomes a primary driver of whole-body insulin resistance.

Given its central role, further investigating the protective effect of skeletal muscle in the context

of diabetes is of high clinical importance. This research moves beyond asking *if* skeletal muscle is associated with protection and instead considers *when* this association is most pronounced. By modelling two distinct clinical thresholds, this study seeks to provide novel evidence that the independent metabolic benefit of skeletal muscle is diabetes stage-specific - that its association with glycaemic control is not uniform over the full glycaemic spectrum.

While the individual roles of adiposity and muscle in glycaemic control across the full range of glycaemia are widely documented, their complex interplay - and particularly the independent stage-specific role of muscle in the face of powerful confounding from fat - is less clear. Understanding these mechanisms will likely lead to clinically relevant knowledge relating to the prevention and treatment of diabetes.

Despite the physiological role of skeletal muscle suggesting a protective association with glycaemic control, there are inconsistencies in the surrounding literature. While several studies have reported that greater muscle mass is linked to a reduced risk of diabetes, others have found no significant association [51] [7]. Interestingly, one study even observed that higher levels of lean mass (which includes skeletal muscle) were positively associated with diabetes risk [35]. These conflicting results may be due to differences in statistical adjustment, particularly a failure to adequately control for the powerful confounding effects of central adiposity, or from modelling the association across the entire glycaemic spectrum as a single, uniform relationship, rather than investigating potential stage-specific differences.

Therefore, this dissertation sought to clarify these discrepancies by examining the independent and stage-specific associations of skeletal muscle with glycaemic regulation. Do these associations remain consistent across the entire glycaemic spectrum, or do they vary by stage? The analysis was based on the hypothesis that, given the established physiological role of muscle in glucose disposal, higher muscle mass would be associated with lower blood glucose levels at all stages.

The structure of this dissertation is as follows: Chapter 3 will detail each stage of the methodology that underpinned the analysis, spanning from outlining fundamental aspects of the data source and variable selection to comprehensive diagnostics of the final statistical models. Next, Chapter 4 will present the results, starting with concise summaries of the final analytical sample and progressing to the key findings that answer the research question. Chapter 5 contains key discussions of the interpretations and implications of the results, also outlining the strengths and limitations of the analysis and suggesting avenues for future research. Finally, Chapter 6 will neatly summarise the complete analysis, stating its contributions to the fields of metabolic epidemiology and data science in health research.

Chapter 3

Methodology

The analytical methods used to investigate the associations of skeletal muscle with glycaemic control involved many sequential stages. The first task was the loading of data from the National Health and Nutrition Examination Survey (NHANES) [27]. Following data loading, meticulous creation and harmonisation of various predictor, outcome, and confounding variables was key. Missing data was appropriately handled using a robust multiple imputation strategy, after which, an iterative modelling journey was conducted to identify the most appropriate statistical models for the goal.

The chosen models were thoroughly diagnosed to ensure they were well-selected and that inferences made using them were accurate. Once the models were known to be well-defined, their coefficients were extracted and examined, they were used to make predictions, and their predictive accuracy was assessed using unseen data.

The following sections will describe each stage of the methodological journey in detail, aiming to instil confidence in, and understanding of, the analysis and findings.

3.1 Data Source: The National Health and Nutrition Examination Survey (NHANES)

The data used for all analyses are from the National Health and Nutrition Examination Survey (NHANES). NHANES is a large-scale health and nutrition examination program conducted by the National Center for Health Statistics - a subsidiary of the CDC.

The program began in 1959 under a different name - The National Health Examination Survey [26]. It was founded to satisfy the National Health Survey Act of 1956, which authorised ongoing surveys “to secure accurate and current statistical information on the amount, distribution, and effects of illness and disability in the United States”.

The overarching goals of NHANES are to deliver reliable data on the prevalence of major health

conditions and their risk factors, monitor changes over time in diseases, behaviours, and environmental exposures, identify new and emerging public health issues, and preserve a nationally representative snapshot of the population’s health and nutritional well-being [26].

The manner in which NHANES collects data is from a combination of personal interviews and questionnaires, standardised health examinations, and laboratory tests. Data such as demographics, socio-economic status, and health behaviours such as smoking history, alcohol consumption, and physical activity were recorded through interviews. Examinations, all performed in Mobile Examination Centres (MECs), were responsible for recording anthropometric and body composition data using dual-energy X-ray absorptiometry (DXA) - the source of the primary muscle mass predictor. Laboratory tests involved the collection of biological specimens such as blood and urine. This was the source of the primary glycaemic control outcome variable.

Approximately 5,000 people take part in NHANES each year, with the data becoming available to the public in biannual cycles. Data from the following cycles have been utilised in this analysis: 2011-12, 2013-14, 2015-16, and 2017-18.

The format of interview questionnaires in NHANES often changes across cycles, causing certain questions to differ in wording or structure. For this reason, a high level of care had to be taken when merging data from each cycle to create the final comprehensive dataset. Techniques such as variable harmonisation and feature engineering were critical to ensure the quality of data was upheld and only meaningful predictors were included in the models. Details of this process can be seen in Section 3.2.4.

Crucial elements of NHANES that needed to be considered were its cross-sectional nature and complex, multi-stage probability sampling design [25]. NHANES is a cross-sectional study, meaning that data from each participant is collected only once at a specific point in time. As a result, analysis of associations between variables is permitted, but inferences of causality are not valid. In addition to its cross-sectional design, accounting for the complex sampling structure enabled appropriate variance estimation, ensuring that the statistics represent the entire U.S. civilian non-institutionalised population. The key features of the NHANES design are as follows:

1. **Strata:** These are groups of Primary Sampling Units (PSUs) that are similar in terms of geography and demographic characteristics, often defined using state-level health indices. Stratification increases the precision of the final estimates.
2. **Primary Sampling Units (PSUs):** These are the clusters, typically counties or groups of counties, sampled within each stratum. PSUs are selected with probabilities proportionate to their size to ensure that subgroups of public health interest are adequately represented. Accounting for clustering is essential for calculating correct standard errors, as individuals within a cluster may be more similar to each other than to the general population.
3. **Sample Weights:** Each participant is assigned a sample weight that can be thought of as counts of the number of individuals that they represent. A participant’s sample weight accounts for their unequal probability of selection (due to oversampling of certain subgroups), as well as adjustments for non-response to the interview and the Mobile Examination Center (MEC) exam (more on this later).

As per the NHANES analytic guidelines, the MEC exam weight was used for all analyses [25]. These are the weights that were used for “the least common denominator”, meaning the “variable of interest that was collected on the smallest number of respondents”. The primary outcome and predictor variables were collected during the physical examination at the MEC. The subset of individuals that participated in the MEC exam was smaller than the group that were interviewed, making the MEC weights the appropriate choice.

All statistical modelling was carried out using the **survey** package in R. This package provides functions that enable a variety of modelling techniques to be used in conjunction with “multi-stage stratified, cluster-sampled, unequally weighted survey samples” [38].

All NHANES data are publicly available on the NCHS website and have been de-identified to maintain the anonymity of the participants. The data can be found alongside comprehensive documentation, including detailed codebooks, survey procedure manuals, and analytical guidelines, the use of which was paramount in ensuring the results of this study were statistically valid.

3.2 Variable Selection and Creation

As previously mentioned, a thorough process of variable selection, creation, and harmonisation across survey cycles was key in this analysis. Scientifically appropriate outcome, predictor, and confounding variables were chosen to successfully isolate the independent role of muscle mass in glycaemic control.

3.2.1 Outcome variable: Glycaemic Control

The outcome variable chosen to represent glycaemic control was glycated haemoglobin (HbA1c) [57].

Although the classification of diabetes has historically relied on single-point measures such as fasting plasma glucose (FPG), clinical guidelines have evolved. HbA1c is now commonly recommended as it provides a more stable and comprehensive assessment by measuring average blood glucose levels over the last 2-3 months [15].

There are many advantages of using HbA1c over single-point measures such as FPG, some of which are as follows:

- FPG levels are subject to a high degree of day-to-day variability due to a variety of factors including acute stress, illness, or changes in exercise [15]. Such variability is not a concern with HbA1c, making it a more stable and reliable biomarker.
- HbA1c can be comparable to roughly thousands of individual measurements of FPG, also capturing postprandial spikes in glucose that are often missed by a simple fasting test, but are significant contributors to overall glycaemic burden [9].
- No fasting is required for retrieving HbA1c measurements, allowing for substantially simpler data collection for large-scale surveys such as NHANES. This also reduces the burden on

participants and eliminates the chance of misclassification due to patients not adhering to fasting protocols.

HbA1c measurement has been standardised worldwide through efforts like the National Glycohemoglobin Standardization Program (NGSP), allowing results from various labs and studies to be comparable (<https://ngsp.org/>).

To test the hypothesis that the association of skeletal muscle with glycaemic control is stage-specific, the continuous HbA1c variable was dichotomised to create two distinct binary outcomes. This approach allows for the separate modelling of the factors associated with the initial onset of dysregulation versus the progression to clinical disease.

The first outcome, representing the “Onset” of elevated risk, was defined using the American Diabetes Association (ADA) threshold for prediabetes [6]. Individuals with HbA1c $< 5.7\%$ were classified as “normal”, while those with values $\geq 5.7\%$ were classified as “elevated”.

The second outcome, representing the “Progression” to diabetes, was defined using the ADA’s diagnostic threshold of 6.5% HbA1c. This separated individuals below the diabetic threshold ($< 6.5\%$) from those with clinical diabetes ($\geq 6.5\%$).

This dual-outcome strategy directly informs the use of two separate binary classification models in the subsequent analysis, allowing for a nuanced investigation of the stage-specific associations. See Section 7.4 for details of how this was coded.

3.2.2 Primary Predictor Variable: Skeletal Muscle Mass

The metric used as the primary predictor and a proxy for skeletal muscle in this study is the appendicular lean mass index (ALMI). This refers to appendicular lean mass (ALM) - muscle mass in the limbs - scaled by height squared. Choosing to only consider muscle in the limbs is key as it allows for greater specificity for skeletal muscle tissue [55]. Including whole body lean mass would introduce muscle in the central organs such as the heart, lungs, and liver. This would result in an inflated measure of skeletal muscle that is not representative of the muscle relevant to glucose uptake or metabolism.

To account for inherent differences in body size, ALM is normalised by height squared to create ALMI. This normalisation is crucial as a taller individual will naturally possess greater appendicular lean mass than a shorter individual, even with a similar physique. Indexing to height squared - a method analogous to the calculation of BMI - allows for valid and meaningful comparisons of muscle mass between individuals of different statures [44].

ALMI is the international standard measurement used for diagnosing sarcopenia, demonstrating its trusted clinical usage [2].

ALMI is not directly reported by NHANES; however, the required data are available, including DXA lean mass measurements (excluding bone mass) for each limb and height measurements. After converting lean mass from grams to kilograms and height from centimetres to metres, these variables were combined as follows to calculate ALMI:

$$\text{ALMI (kg/m}^2\text{)} = (\text{Left leg lean mass (kg)} + \text{Right leg lean mass (kg)} + \text{Left arm lean mass (kg)} + \text{Right arm lean mass (kg)}) / \text{Height}^2 \text{ (m)}$$

This conversion was done after multiple imputation of the appropriate variables in each imputed dataset (see Section 7.4). Doing this prior to imputation would not have been optimal. The multiple imputation algorithm works best when it has the largest amount of information to learn from. Prematurely combining the lean mass measurements of all four limbs into ALMI would reduce the information that the imputation is able to learn from, likely reducing its accuracy. In addition, calculating ALMI before imputation would result in the loss of observations in the dataset. This would be due to a missing value in any of the four limb measurements, or height, resulting in NA when combined. Forming the ALMI variable after the imputation algorithm is key in maintaining the highest possible accuracy and size of the final dataset.

3.2.3 Selection of Simple Confounding Variables

To isolate the independent effect of skeletal muscle on glycaemic control, it is essential to control for potential confounding variables. Failure to do so can lead to biased estimates and spurious correlations. Confounding variables are those that are associated with both the outcome and primary predictor, but are not on the causal pathway between them [33].

Choosing what confounding variables to include in the models was done based on the backing of the surrounding literature. Some confounders were simple to obtain and required no extensive harmonisation. These variables, and the reasons for including them, are outlined in this section. See Section 3.2.4 for details on the inclusion of those variables that required substantial pre-processing.

Key demographic confounders included in the analysis are age, gender, and race. Age is inversely related to muscle mass, which typically peaks in young adulthood and naturally declines thereafter - a process known as sarcopenia [50]. Body composition, including skeletal muscle, also differs between males and females, and these differences extend across different ethnic backgrounds.

Furthermore, increasing age is strongly associated with worsening glycaemic control due to insulin sensitivity decreasing with age [11]. Sex hormones have been shown to have a strong influence on insulin sensitivity, resulting in differing risk profiles for metabolic disease between men and women, with women appearing to have lower insulin resistance than men [19]. Finally, there is a highly documented disparity in the prevalence of type 2 diabetes between those of differing ethnicities [11].

These reasons provide strong justification for the inclusion of age, gender, and race as confounding variables in the statistical models. Each variable is readily available in each cycle of NHANES and was directly extracted to be included in the final analytical sample.

To comprehensively account for the confounding effects of adiposity, key body composition variables - body fat percentage (BFP) and waist circumference (WC) - were included in the models. As demonstrated by research directly comparing these metrics, both are significant predictors of type 2 diabetes, with central adiposity often showing a particularly strong association with the

incidence of the disease [22]. Therefore, including both is essential to fully disentangle their effects from the independent contribution of muscle mass. Furthermore, adiposity is inversely associated with muscle mass, and failure to account for this relationship can confound the effects of muscle mass on health outcomes. For example, even those with BMI in the normal ranges, individuals that have low muscle mass may have higher body fat percentages and increased mortality risks [1].

Another confounding variable that was considered in the analysis was diet quality due to it being a primary determinant of glycaemic control and having strong association with skeletal muscle mass in the body [18] [13]. NHANES collects substantial data regarding diet in the form of “24-hour dietary recalls” [28]. Since the 2002–2003 survey cycle, each participant has completed two 24-hour dietary recalls, the second occurring 3 to 10 days after the first. Each recall collected detailed information on foods consumed and total nutrient intakes.

There is no easy way to manually leverage the complex NHANES dietary recall data to engineer a single variable to include in the models, therefore, the R package `heiscore` was used to seamlessly extract what is known as Healthy Eating Index (HEI) scores for each participant of the required NHANES survey cycles. The HEI is a tool used to evaluate diet quality by measuring how closely a group of foods adhere to the core recommendations and dietary patterns outlined in the Dietary Guidelines for Americans [58]. A HEI score is a number from 1 to 100, with numbers closer to 100 indicating a healthier diet.

The final simple confounding variable that was been included in the statistical models was the ratio of family income to poverty (PIR), which is used as a measure of socio-economic status (SES). SES is a well-established social determinant of health, strongly associated with health behaviours, access to healthcare, and the risk of developing type 2 diabetes [32] [54]. It is a multifaceted construct that is frequently utilised in epidemiological research through indicators such as educational attainment and occupational status [46]. These indicators may be problematic by attaching excessive significance to education and employment, while failing to acknowledge access to, and affordability of healthcare.

This concern is rectified by the use of PIR, which represents household income relative to the appropriate poverty line. The poverty line or threshold is the minimum income level deemed necessary for a family to meet basic needs like food, housing, and clothing, and family income is defined as the total income of all members of a given household or family. A PIR of 1 means the family’s income is equal to their poverty threshold, less than one implies they are economically disadvantaged, and greater than one implies they are financially secure, with higher magnitudes of PIR indicating further distance from the poverty line.

3.2.4 Data Processing and Harmonisation of Lifestyle Confounders

Section 3.2.3 outlined the selection of the confounding variables that did not require significant adjustment to be compiled in the final dataset. As previously mentioned, some confounders did require extensive data processing due to reasons such as structural missingness as a result of survey skip patterns and changes to interview structure over different survey cycles.

This section will provide justification for why these remaining confounders were chosen, and it will detail the meticulous wrangling that was needed to form the final comprehensive dataset prior to multiple imputation, including the harmonisation of these more complex confounders over each survey cycle of NHANES, and the techniques used to handle structural missingness. All the required R code for these tasks can be seen in Section 7.3.

3.2.4.1 Combining NHANES Survey Cycles

The initial task was to combine each relevant dataset from each cycle of NHANES from 2011 to 2018, keeping only the variables required for the analysis. These were each of the aforementioned variables in Section 3.2, an individual ID, and the variables that were needed to engineer lifestyle confounders of physical activity, alcohol consumption, cigarette smoking, and daily hours of sleep. The details of such feature engineering will be outlined later in this section.

Each required dataset within NHANES was obtained using the `nhanesA` package in R. This package provides a simple function `nhanes` which was used to load the relevant data from each survey cycle into R. This process can be seen in Section 7.2, along with the use of the `score` function from the aforementioned `heiscore` package to obtain the HEI scores for each participant.

Inner join was used, with the ID variable as the key, to combine each dataset within each survey cycle using a custom function `multi_inner_join`. This function took a list of data frames as an input and returned the result of combining them with inner join. The use of inner join was crucial in creating a valid analytical sample prior to imputation. Not all individuals who participated in the interview stage also took part in the examination phase, and vice versa. Merging datasets with an inner join ensures that only participants who completed both the interview and examination stages are included in the final dataset. Without this step, data not missing at random would need to be imputed, leading to a substantially larger but severely biased dataset. Each survey cycle’s combined data frame was given a new identifier column to label which survey cycle its observations were from. This variable was key in the variable harmonisation phase.

Next, the complete merged datasets for each survey cycle were simplified by removing every variable within them that was not required. Following this, a new large dataset was formed by binding each survey cycle’s dataset by row. This dataset contained 12,178 rows, comprising of all participants that took part in both the interview and examination phases in one of any NHANES survey from 2011 to 2018. Due to the cross-sectional nature of NHANES, no individual took part in multiple survey cycles.

3.2.4.2 Physical Activity

Following the creation of this large dataset was the handling of structural missingness and harmonisation of lifestyle confounders, the first of which being physical activity, which is well known to be strongly associated with skeletal muscle mass and glycaemic control [12] [41]. The physical activity variable used for modelling combined total minutes of “moderate” and “vigorous” work-related and recreational activity in a typical day (MVPA). Definitions of these activity levels are provided in the NHANES physical activity questionnaire codebooks (2017-18 [24]).

The logic of the physical activity questionnaires in NHANES was vital to consider when forming a final physical activity variable to be used for statistical modelling. Before each participant was asked to report the number of minutes of each type of physical activity of either intensity they undertake in a typical day, they were first asked if they engage in any at all for at least 10 minutes continuously in a typical week. These responses were stored in the following variables for moderate/vigorous activity during work or recreationally, respectively: PAQ665/PAQ650 and PAQ620/PAQ605. If the participant gave the answer “Yes”, they were also asked how many minutes of the appropriate level of activity they carry out in a typical day - PAD675/PAD660 for moderate/vigorous recreational activity and equally PAD630/PAD615 for work-related activity.

This structure introduces structural missingness. Any time a participant answered “No” to the initial gatekeeper question, they were not asked the follow up question, resulting in the skipped response being coded as NA in the data. Of course, this data is not “missing”. An initial answer of “No” should correspond to 0 minutes of the appropriate level of physical activity in a typical day. This issue was resolved by recoding the variables as needed (see Section 7.3). Very few individuals also responded with “Don’t know” when asked the follow-up question regarding minutes of activity. This was coded as 9999 in the dataset and was handled by setting 9999 values to NA so that they could be imputed. Handling the outlined structural missingness resulted in the proportion of missing data for PAD675, PAD660, PAD630, and PAD615 reducing from 56%, 69%, 60%, and 77%, respectively, to 0.09%, 0.04%, 0.3%, and 0.2%, emphasising the importance of this data processing. The final physical activity variable was formed by summing PAD675, PAD660, PAD630, and PAD615.

Some values of this final physical activity variable were implausibly high. It is not possible to discern why this was the case, but a likely estimate would be due to questions being misunderstood by participants. It was made clear in the questions of the physical activity questionnaires to exclude activity included in the answers of previous questions. This instruction may have not been followed, potentially leading to cases of extreme combined activity values.

Values above a threshold of 960 minutes were assumed to be missing and set to NA to be imputed. The choice of this specific threshold was informed by the UK Biobank [8].

3.2.4.3 Alcohol Consumption

The next lifestyle confounder to be included was alcohol consumption due to prior evidence showing that alcohol affects both muscle composition and glycaemic regulation [49] [30]. The alcohol consumption questionnaire from the NHANES interview stage maintained the same structure for the relevant questions from 2011 to 2016, but underwent alterations for the 2017-2018 cycle. This necessitated a complex multi-stage harmonisation process that resulted in a single, robust 4-level categorical variable (`Alcohol_Status`) distinguishing never drinkers, former drinkers, moderate drinkers, and heavy drinkers. Thresholds outlined by the U.S. government’s “Dietary Guidelines for Americans” were used to classify drinkers as “moderate” or “heavy” [4].

Differentiating those who have never consumed alcohol and those who are former drinkers is key in handling the “sick-quitter” effect. The sick-quitter effect refers to individuals who stop drinking alcohol because of health problems [31]. Although they are often categorised as non-drinkers

in health data, they may have already experienced alcohol-related harm before quitting. When studies compare drinkers to non-drinkers and the non-drinker group includes former drinkers with existing health issues, it can distort the results. This may make the risks of alcohol seem lower than they are, or even suggest that light drinking offers health benefits. The construction of the **Alcohol_Status** variable handles this appropriately, substantially improving the quality of the final variable and more effectively controlling for the confounding effects of alcohol.

A categorical approach offers key advantages over using a continuous variable of alcoholic drink consumption. The sick-quitter effect would be ignored using a continuous approach, resulting in both “Never Drinkers” and “Former Drinkers” being coded as 0, and it captures non-linear risk, which avoids the invalid assumption of a linear dose-response relationship.

The harmonisation process involved the following steps:

1. Harmonisation of Drinking Frequency:

This stage involved creating a consistent measure of how often participants drank alcohol. For the 2011-2016 cycles, this was recorded as a numerical variable directly measuring the number of drinking days in the last year (**ALQ120Q**). However, for the 2017-2018 cycle, this was replaced with a categorical variable (**ALQ121**). To create a single, continuous variable that was comparable across all cycles (**Drinking_Days_Per_Year**), the categorical **ALQ121** was converted into an approximate number of days per year (e.g., “Every day” was converted to 365, “Once a week” to 52, etc.). The new **Drinking_Days_Per_Year** variable was simply the same as **ALQ120Q** for the responses from 2011-2016. Codes for “Refused” and “Don’t know” responses were assigned to **NA** across all cycles to be imputed.

2. Cleaning of Drinking Quantity

The variable for the quantity of drinks per drinking day (**ALQ130**) was cleaned to handle “Refused” and “Don’t know” codes in the same way as in Step 1. In the original data, values of **ALQ130** were coded as **NA** for those whose answers to **ALQ120Q** or **ALQ121** were 0 or “Never in the last year” respectively. Of course, this was not true missing data, but rather should be coded as 0. This was fixed by assigning a new clean **ALQ130** variable that was 0 whenever **Drinking_Days_Per_Year** was 0.

3. Calculation of Average Daily Intake

The harmonised frequency and cleaned quantity variables were then used to calculate a final, continuous variable denoted by **AvgDailyDrinks** that measured the average number of daily alcoholic drinks consumed by each participant over the year prior to their survey cycle.

4. Harmonisation of Lifetime Drinking Status

This step involved creating a consistent flag to identify lifelong abstainers, enabling clear distinction between “never drinkers” and “former drinkers”. The relevant questions differed by cycle: **ALQ110** (2011–2016) asked whether participants had consumed at least 12 alcoholic drinks in their lifetime, while **ALQ111** (2017–2018) simply asked if they had ever consumed an alcoholic drink. Additionally, before answering **ALQ110**, participants in 2011–2016 were

first asked whether they had consumed at least 12 drinks in the past year. A “Yes” response skipped ALQ110, leaving it coded as NA when the correct value should have been “Yes”. This structural missingness was addressed appropriately, and “Refused” and “Don’t Know” responses were recoded to NA for later imputation. Finally, a harmonised variable (LifetimeDrinkerFlag) was created using the cleaned ALQ110 for 2011–2016 and ALQ111 for 2017–2018. The distinction between having at least 12 drinks versus a single drink in a lifetime could not be preserved and was treated as equivalent.

5. Final Variable Correction and Selection

Due to survey skip patterns, there were instances where AvgDailyDrinks was coded as NA where LifetimeDrinkerFlag was “Never”. This was simple to resolve by setting AvgDailyDrinks to 0 in each of these cases. Following this, the LifetimeDrinkerFlag variable was converted to a factor with the levels “Never” and “Ever”, and the relevant final columns of the dataset were selected, allowing AvgDailyDrinks and LifetimeDrinkerFlag to be passed into the multiple imputation algorithm.

6. Post Imputation Variable Creation

Finally, after multiple imputation, AvgDailyDrinks and LifetimeDrinkerFlag were used to create the final 4-level Alcohol_Status variable. This process can be seen in [Section 7.4](#).

3.2.4.4 Smoking Status

Smoking cigarettes has been shown to negatively impact both muscle mass and blood sugar levels [61] [60]. For this reason, there was strong justification to include smoking status as a confounding variable in the statistical models for this analysis.

The goal was to create a clean categorical Smoking_Status variable using the results of two key questions in the smoking interview from NHANES, namely SMQ020 (“Have you smoked at least 100 cigarettes in your life?”) and SMQ040 (“Do you now smoke cigarettes?”). This allowed for Smoking_Status to be a 3-level factor of “Never Smoker”, “Former Smoker”, and “Current Smoker”, handling the sick-quitter phenomenon effectively. As with the alcohol consumption variable, a categorical approach was also chosen here due to handling of the sick-quitter effect and not assuming a linear dose-response relationship. Unfortunately, there was no question that asked participants whether they had ever smoked in their life, meaning that “Never Smokers” were classified based on if they answered “No” to SMQ020.

The process of engineering this variable was largely simple due to the structure of the smoking questionnaire not changing over different survey cycles. The first step was to assign those who answered “No” to SMQ020 as “Never Smoker”, as previously mentioned. Next, responses to SMQ040 were used to distinguish between those who were former smokers and those who currently smoke. If the response to SMQ020 or SMQ040 was “Refused” or “Don’t know”, Smoking_Status was assigned as NA to be imputed. Finally, the factor levels of Smoking_Status were assigned, and the relevant columns of the data frame were chosen to lead into further data processing and imputation.

3.2.4.5 Daily Hours of Sleep

The final confounding variable to be engineered and included in the statistical modelling was average daily hours of sleep, which also has been shown in the surrounding literature to be associated with muscle mass and glycated haemoglobin [3] [45]. A minor harmonisation challenge arose as the variable name for the relevant question in the NHANES questionnaire changed across survey cycles. The question was recorded under the variable `SLD010H` in the 2011-2014 cycles and under `SLD012` in the 2015-2018 cycles. These were combined into a single, consistent `AvgNightlySleep` variable for the entire eight-year study period. The final variable was then cleaned to remove any special codes, and the final dataset was selected, removing any unnecessary columns.

3.3 Imputing Missing Data

Following the extensive data processing and harmonisation outlined in the previous section, which addressed all identifiable structural missingness, a final analytical dataset was formed. Within this dataset, approximately 5% of the total data points remained missing across all variables. A naive approach to handling this issue - listwise deletion - was deemed inappropriate for two primary reasons. Although only 5% of the data was missing, conducting listwise deletion reduced the sample size by 35%, leading to a significant loss of statistical power. Secondly, this method is only valid under the strict and implausible assumption that the data are Missing Completely At Random (MCAR) [10]. It is far more likely that in a complex health survey like NHANES, the data are Missing at Random (MAR), meaning the probability of a value being missing is related to other observed variables. Using listwise deletion on MAR data is known to introduce bias into the results.

For these reasons, a more sophisticated strategy was required - multiple imputation. This method works by replacing the missing values with plausible estimates drawn from a distribution that is informed by all of the other variables in the dataset. The algorithm takes an incomplete dataset as input, and returns multiple complete datasets that are formed this way. In this analysis, multiple imputation was conducted using the `Amelia` package in R, with a generous number of 30 imputations to ensure statistical power and minimal bias.

Before conducting the imputation, it was beneficial to define physiologically informed bounds of each variable to be imputed. This prevented the imputation algorithm from generating impossible values such as a negative waist circumference or age of 200. Parallel processing was also utilised when running the imputation algorithm. This allowed for the use of multiple CPU cores, vastly speeding up computation times. After imputation, there was a minor data processing task which involved calculating the ALMI primary predictor, creating the `Alcohol_Status` variable using the now complete `AvgDailyDrinks` and `LifetimeDrinkerFlag` variables, creating the `at_risk_or_worse` and `is_diabetic` outcome variables for the onset and progression models respectively, and removing variables that were no longer needed. This was done for all 30 imputed datasets. See Section 7.4 for details on these processes.

3.4 Descriptive Analysis and Exploratory Visualisation

Concise summaries and visualisations of the final analytical sample were then produced prior to model fitting. This was key to ensure robust understanding of the data before conducting the analysis.

The first stage was to create a summary table that displayed the mean and standard deviation of all continuous variables, and the count and proportion of all categorical variables in the dataset. However, to ensure accuracy, it was important to consider the survey design and each of the 30 imputed datasets for this process.

A custom function was used to obtain the relevant summary statistics from a single imputation, being sure to account for survey design using functions such as `svymean` and `svyvar` for survey-weighted means and variances respectively. This was applied to all imputations, and the results were stacked using the `map_df` function. The stacked results were then averaged to produce the final table.

To supplement the descriptive statistics table, visualisations were created using `ggplot2` to show the distributions of the primary predictor (ALMI) and the continuous form of the outcome variables (HbA1c), as well as their unadjusted relationship. A custom theme was used to allow for visually appealing and concise plots, which was used for all other `ggplot2` graphics in the analysis.

The distribution of ALMI and HbA1c were shown using density plots. Density curves from each of the 30 imputations were superimposed over each other, allowing the uncertainty introduced by the imputation algorithm to be visualised. For visual clarity, the scatter plot of the relationship between the variables used only imputation 1.

3.5 Statistical Model Fitting

Statistical modelling was carried out following the completion of data processing and multiple imputation. As described in Section 3.2.1, this was achieved using a dual-model framework:

- Onset models to investigate the transition from normal to elevated HbA1c.
- Progression models to investigate the transition from a non-diabetic to a clinical diabetic state.

The use of the `survey` R package was key in fitting models that respected the survey design of the NHANES data. Prior to model fitting, the survey design of each of the 30 imputed datasets was specified using `svydesign`. Due to fitting models that respect the survey structure, the survey design object was supplied to the fitting functions rather than the raw datasets.

3.5.1 Modelling Strategy

Prior to the dual-model approach, the initial modelling strategy was to use survey weighted ordinal logistic regression with the R function `svyolr`. This method involved fitting models with a single

categorical outcome variable rather than the dual-outcome approach with the outcome variables of `at_risk_or_worse` and `is_diabetic`. The outcome variable was a 3-level factor of the following categories:

1. Normal HbA1c ($< 5.7\%$)
2. Prediabetic HbA1c ($\geq 5.7\% \ \& \ < 6.5\%$)
3. Diabetic HbA1c ($\geq 6.5\%$)

This form of model attempted to predict the cumulative log odds of being at or below a certain step of the ordered outcome and how the predictors affect it. The key assumption that is associated with ordinal regression is that of proportional odds. The assumption is that the effect of any given predictor is the same across all the different thresholds of the outcome variable. This means that the effect of having more muscle (ALMI) on the odds of moving from “Normal” to “Prediabetic” should be the same as moving from “Prediabetic to” “Diabetic”.

For the ordinal regression approach to be valid, the proportional odds assumption must be met. A formal test was conducted by fitting two separate binary logistic regression models: one comparing “Normal” vs. “Prediabetic” or “Diabetic” and a second comparing “Normal” or “Prediabetic” vs. “Diabetic”. The results of this test revealed that the coefficients for key predictors, including ALMI, were significantly different between these two models. This violation was not merely a statistical issue; it provided direct evidence that the association between skeletal muscle and glycaemia is stage-specific. Therefore, the ordinal approach was abandoned as it could not adequately capture this complexity, necessitating the adoption of the final dual-model framework.

The first application of the final binary classification modelling framework was using survey weighted generalised linear models with a quasi-Poisson family and log link function through the `svyglm` function in R. This approach is often favoured in epidemiological research as it allows for the direct estimation of prevalence ratios (or risk ratios), which can be more intuitive to interpret than odds ratios [62]. However, goodness-of-fit diagnostic tests confirmed that onset models fit using the quasi-Poisson assumption were not well-defined, and did not fit the data well. The goodness-of-fit test used was the Archer-Lemeshow test - a specialised diagnostic tool for assessing the fit of binary classification models fitted using survey weighted data [5]. More detail on this diagnostics process can be seen in Section 3.6, in which the final models were thoroughly tested.

Consequently, the quasi-Poisson approach was discarded in favour of the quasi-binomial approach, as this provided a stable, well-fitting model for both outcomes - a result later confirmed by goodness-of-fit testing. The results are therefore presented as odds ratios.

The complex multi-stage probability sampling design employed by NHANES, in particular the clustering of observations within PSUs, leads to a violation of the assumption of independence among observations inherent in standard GLMs. This intra-cluster correlation is known to produce apparent overdispersion, which, if left unaddressed, would result in badly defined standard errors [37]. The model fitting procedure followed by `svyglm` handles this overdispersion by using a design-based variance estimator (more on this later in this section). The use of “quasi” versions of the model family objects is used to simply avoid warnings in R regarding non-integer responses that

occur due to survey weights - not to handle overdispersion, as is common when using unweighted GLMs. This is recommended in the documentation of the `survey` package.

The mathematical definition of the final models is specified as follows:

Let Y_i equal the binary outcome for an individual i where $Y_i = 1$ if the individual has the condition (prediabetes for the Onset outcome and diabetes for the Progression outcome), and $Y_i = 0$ if they do not. Let π_i equal the probability that $Y_i = 1$ - the mean. Unlike in standard binomial regression, the model does not assume a binomial likelihood. The mean structure is the same as in an unweighted GLM, while variance for inference is obtained using survey-weighted methods that account for the complex sampling design.

The mean is modelled using a logit link function that ensures that it is of the correct scale - a proportion between 0 and 1. This links the mean to a linear combination of the predictor variables and is defined as follows:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \dots + \beta_k X_{k,i}$$

where β_0 is the intercept, $\beta_1, \dots, \beta_k$ are the regression coefficients for the k predictor variables, and $X_{1,i}, \dots, X_{k,i}$ are the values of the predictor variables for the i th individual.

The coefficients $\beta_1, \dots, \beta_k$ are estimated using weighted estimating equations as outlined by Thomas Lumley and Alastair Scott in their 2017 article regarding fitting regression models to survey data [39].

The variance of the estimated coefficients is computed using Horvitz–Thompson-type standard errors, which is a generalisation of the robust sandwich estimation procedure outlined by Zou in his 2004 article outlining the aforementioned Poisson-based approach for estimating risk ratios in epidemiological studies such as this [62] [39]. This is the design-based variance estimator. Unlike in an unweighted quasi-binomial GLM, this approach does not include scaling the variance by an estimated dispersion parameter, while still providing appropriate standard errors, confidence intervals, and p-values under the complex sampling design.

3.5.2 Stepwise Model Building

To understand the influence of different confounders and select a final model for both the Onset and Progression outcomes, a hierarchical model-building approach was employed.

This process involved fitting a sequence of ten nested models, where logical blocks of covariates were added at each step. The initial model of the sequence was crude and only contained the primary predictor (ALMI). Each successive model progressively adjusted for demographics, body composition, socio-economic status, and finally, all lifestyle factors.

A critical aspect of this process was that it was applied across the multiply imputed data. Each of the ten models in the sequence were fitted to all 30 imputed datasets, and the results were pooled using Rubin’s Rules by using the `MIcombine` function in R. This resulted in a single, robust set of coefficient estimates for each individual model definition.

This systematic model building strategy functioned as a sensitivity analysis, demonstrating how the ALMI coefficient responded to the inclusion of each confounding variable. To formally investigate the statistical mechanism behind these changes, particularly the multicollinearity between the three body composition variables, survey-weighted Variance Inflation Factor (VIF) values were calculated at key steps of the model-building process.

The final model for the Progression outcome was chosen to contain only the confounders that, once added, changed the estimation of the ALMI coefficient by 10% or more [59]. This provided a data-driven justification for selecting a parsimonious model that was both well-fitting and did not possess needless complexity. For the Onset outcome, a different modelling strategy was planned, with the aim of providing a comprehensive adjustment for all potential confounders. Details of this approach, along with the justification for why it differed from the Progression outcome, are provided in Section 4.

3.6 Model Diagnostics

Following the selection of the final models, a suite of diagnostics tests were carried out to ensure each model was well-defined, robust, and suitable to make inferences from. Section 7.7 contains the code needed for the processes outlined in this section.

The first stage of this process was conducting Archer-Lemeshow goodness-of-fit tests. The test was run on each of 30 imputed datasets for a given model, and the results were pooled appropriately. The Archer-Lemeshow test outputs a single p-value to indicate goodness-of-fit. It is statistically invalid to simply average p-values across imputed datasets to obtain a single result for each model, in a similar way that combining model coefficients across imputations must be done using Rubin’s rules to account for the within- and between-imputation variances. Therefore, a specific pooling procedure based on the test’s F-statistic was required.

For a given model, the F-statistic of the test for each of the 30 imputations was stored in a list. This list, along with the appropriate degrees of freedom, was then supplied to the `micombine.F` function, which viably pooled the test statistics and returned a single p-value that was used to assess the fit of a given pooled model. In this case, p-values larger than 0.05 indicated a good fit.

To compare the relative explanatory power of the nested models, McFadden’s Pseudo R^2 values were calculated. While not a direct analogue to R^2 in linear regression, it serves as a valuable metric for assessing the proportional reduction in model deviance, with values between 0.2 and 0.4 often considered to indicate an excellent fit [29].

A key technical consideration is that survey-weighted models fit using `svyglm` do not have a true likelihood and, consequently, no true deviance. The `fit.svyglm` function used in this analysis calculates the Pseudo R^2 by applying the un-weighted deviance formula to the weighted model formula [53]. Therefore, the resulting Pseudo R^2 should be interpreted as a heuristic for comparing relative model fit, rather than as an absolute measure of variance explained.

McFadden’s Pseudo R^2 was calculated for all 30 imputed datasets for each model, and the mean

was taken to represent the final statistic for each pooled model. Simply averaging the R^2 values is argued to be more appropriate than pooling using methods such as Rubin’s rules [21].

To further assess model validity, visualisations of binned residual plots were used. This was only done using the final Onset and Progression models chosen using the methods outlined in Section 3.5.2.

Residual diagnostics are key in model evaluation to ensure that models are well-specified, and to discern whether higher order terms of the covariates are required. Standard residual plots are not useful in a logistic regression context due to the binary outcome variable. Binned residual plots overcome this by grouping observations into bins based on their predicted values and plotting the average residual vs average fitted value in each bin [20]. Any clear systematic patterns, or too many points outside the ± 2 standard error bounds, would indicate potential misclassification and warrant further testing. It is expected that around 95% of points should be contained within the ± 2 standard error bounds.

Linear predictor vs binned residuals plots were produced to assess overall model fit, and component-wise residual plots were produced to more accurately validate the assumption of linear continuous covariates. Due to the multiple imputation framework, linear predictor binned residual plots were produced for each of the 30 imputed datasets for the required two models. These plots were compared to identify any differences between imputations, from which it was discerned that none were significant. Therefore, component-wise residual plots were only formed using the first imputed dataset.

As a final validation check, it was useful to investigate any significant outliers in the data to ensure that model conclusions were robust and not disproportionately influenced by a small number of unusual observations. The primary diagnostics tool for this was Cook’s distance, which was calculated for the final models using the R function `svyCooksD` to correctly account for the complex survey design.

The first stage of this test was to identify which observations were the most influential. This was done by finding those with the top 1% Cook’s distance values from the final Onset and Progression models respectively. Due to residual diagnostics indicating no significant variation between imputations, this process was done only on the models fit to imputation 1. Next, these models were refit using redefined survey design objects formed using the first imputed dataset with these points removed. The coefficients from the original and refitted models were then compared to assess the stability of the findings.

3.7 Model Prediction

To aid in the interpretation of the primary identified associations, predicted probabilities were generated and plotted. The goal of this process was to clearly illustrate the shape and magnitude of the key associations of skeletal muscle and glycaemic control that were uncovered during the analysis, along with their surrounding uncertainty. Predictions were generated using the final parsimonious Progression model, allowing for a visual representation of the associations of ALMI

and the probability of becoming diabetic. This was done separately for males and females due to previously discussed well-documented differences in hormonal profiles and body composition between the sexes.

Predictions were initially made in the logit scale, and then converted back to the probability scale after the calculation of the confidence intervals. This was done to avoid incorrect calculations of negative confidence intervals.

The initial step was to create a new dataset containing the covariate values for prediction. A sequence of evenly spaced ALMI values was selected from the observed minimum to maximum, continuous covariates such as age and body fat percentage were fixed at their means, and categorical variables, such as race, were set to their mode.

To fully account for uncertainty due to missing data, continuous variables other than age were fixed at their means across all 30 imputed datasets. Age and race contained no missing values and were therefore assigned their values from the first imputation. Predictions were then generated across all 30 imputed datasets, with the final point estimate obtained as the mean of these predictions. Confidence intervals were constructed by combining the within- and between-imputation variances using Rubin's Rules, resulting in a total variance for interval estimation. The resulting mean predictions and confidence intervals were visualised using `ggplot2`. The code for the processes outlined in this section can be seen in Section 7.9.

3.8 Assessment of Predictive Accuracy (Cross-Validation)

The final stage of this analysis was to evaluate the predictive performance of the final parsimonious Progression model using k-fold cross-validation. While the primary objective of this study was explanatory rather than the development of a classification tool, this step provided an important assessment of the model's generalisability by quantifying its performance on data not used in model fitting.

Due to the computational complexity of running the procedure across all 30 imputed datasets, the cross-validation was conducted on the first imputed dataset as a representative sample. As residual diagnostics confirmed no significant differences between imputations, this is a valid approach.

Although automated R packages for carrying out survey-weighted cross-validation are available, a custom algorithm was used to allow for improved flexibility and specific control of the process. The final validation involved 5-fold cross-validation repeated 10 times to ensure robust results and no dependency on the random allocation of the 5 folds.

This procedure was only carried out using the Progression model outcome for reasons made clear in Section 4.6.

For each of the 10 repeats, the following stages were carried out for each fold:

1. The dataset was partitioned into training and testing portions, with 80% allocated for training, and 20% for testing.

2. A survey design object was created using the training dataset, and was then used to fit a survey weighted logistic regression model using the same model specification as the final parsimonious Progression model.
3. The model was then used to generate probability predictions for the training data. An optimal classification threshold was determined by calculating a survey weighted Receiver Operating Classification (ROC) curve on the training predictions and selecting the threshold that maximised Youden’s J statistic. This method ensures the greatest distance between sensitivity and the false positive rate, resulting in an optimal balance between sensitivity and specificity [23].
4. The model trained in Step 2 and optimal threshold determined in Step 3 were applied to the test data, classifying observations as either diabetic or non-diabetic.
5. Based on the results, performance metrics were calculated on the test set. The primary metric was the Area Under the Curve (AUC), which quantified the model’s ability to distinguish between the two classes on a scale from 0.5 to 1. An AUC of 0.5 means a model has no discriminatory power, whereas an AUC of 1 indicates perfect classification. AUC values of above 0.8 are considered to indicate that a model has clinical utility [16]. Further measures of performance such as sensitivity, specificity, and overall accuracy were also stored. The process of calculating performance metrics required careful consideration of the survey design using functions such as `WeightedAUC` for survey weighted AUCs, `svymean` for calculation of weighted errors, and `svytable` for weighted confusions matrices.

The mean performance metrics from each repeat were averaged to produce the final estimates of the model’s predictive performance. To further aid in the understanding of the procedure, the ROC curve from the final iteration of the cross-validation was plotted and labelled with the optimal threshold and AUC.

Chapter 4

Results

This chapter will present the findings of the analytical processes outlined in the methodology. It will begin by providing a broad summary of the final analytical sample formed by the diligent data processing discussed in Section 3.2, and then present the results of the hierarchical model building procedure that led to the choice of the final models. Inferences of the final models will then be presented, detailing the key findings that they have made possible. Finally, the results of model-based prediction and a thorough assessment of the predictive accuracy of the final parsimonious Progression model are shown.

4.1 Descriptive Statistics of the Analytical Sample

Once the raw data had been fully processed and all missing values had been imputed, the final analytical sample consisted of 12,178 participants between the ages of 18 and 59. The pooled, survey-weighted characteristics of the sample are presented in Table 1.

The mean age of participants in the sample was 39 years ($SD = 12.3$), indicating a wide age distribution representative of the adult US population. 51.7% of the sample were female. The population was predominantly Non-Hispanic White (61.9%), with the second most populous group being Non-Hispanic Black (12.3%), and the remaining groups each comprising under 10% of the sample.

The mean value of the primary predictor (ALMI) was 8.0 kg/m^2 ($SD = 1.7$), and the mean HbA1c was 5.5% ($SD = 0.9$). The distribution of the outcome variables for the Onset and Progression models showed that 76% of the population were in the normal healthy range of blood glucose levels, leaving 24% in the elevated range ($\geq 5.7\%$). In addition, 94% had HbA1c below the diabetic threshold, meaning 6% were in the range of clinical diabetes ($\geq 6.5\%$).

Table 1: Pooled Characteristics of the Final Analytical Sample

Characteristic	Value
Age (years)	39.0 (12.3)
ALMI (kg/m ²)	8.0 (1.7)
HbA1c (%)	5.5 (0.9)
Body Fat Percentage (%)	33.5 (8.7)
Waist Circumference (cm)	98.7 (17.4)
Poverty Income Ratio	2.9 (1.7)
Physical Activity (mins/day)	164.4 (196.6)
Healthy Eating Index	52.6 (13.5)
Average Nightly Sleep (hours)	7.3 (1.4)
Glycaemic Status (Onset)	
Normal HbA1c	9252 (76.0%)
Elevated HbA1c	2926 (24.0%)
Diabetes Status (Progression)	
Non-Diabetic	11445 (94.0%)
Diabetic	733 (6.0%)
Gender	
Male	5886 (48.3%)
Female	6292 (51.7%)
Race	
Mexican American	1209 (9.9%)
Hispanic	833 (6.8%)
Non-Hispanic White	7543 (61.9%)
Non-Hispanic Black	1493 (12.3%)
Non-Hispanic Asian	636 (5.2%)
Other Race	465 (3.8%)
Alcohol Consumption	
Never Drinker	1395 (11.5%)
Former Drinker	1291 (10.6%)
Moderate Drinker	9230 (75.8%)
Heavy Drinker	261 (2.1%)
Smoking Status	
Never Smoker	7369 (60.5%)
Former Smoker	2348 (19.3%)
Current Smoker	2460 (20.2%)

Note:

ALMI: Appendicular Lean Mass Index; HbA1c: Haemoglobin A1c.

* Continuous variables presented as mean (SD), categorical variables presented as n (%).

Body composition characteristics showed considerable variability, with a mean body fat percentage of 33.5% (SD = 8.7) and a mean waist circumference of 98.7 cm (SD = 17.4). The mean ratio of family income to poverty was 2.9 (SD = 1.7), reflecting a wide spectrum of socio-economic conditions across the sample.

Lifestyle factors also varied substantially. The mean Healthy Eating Index score was 52.6 (SD = 13.5), and the average nightly sleep was 7.2 hours (SD = 1.4). Of particular note, recreational physical activity levels were highly skewed. While the mean was 55 minutes per day, the large standard deviation of 75.1 minutes indicates that many participants reported little to no activity, while a smaller number of highly active individuals significantly influenced the average.

The vast majority of the cohort were classified as moderate drinkers, with this group containing 75.8% of the sample. Only 2.2% of participants were heavy drinkers, 10.6% were former drinkers, and 11.5% had never consumed an alcoholic beverage. The significant number of people in the

“Former Drinker” category, namely 1291, highlights the importance of handling the sick-quitter bias. In contrast, the majority of individuals had never smoked cigarettes, with the “Never Smoker” category consisting of 60.6% of the population. 19.3% of people were former smokers, leaving 20.2% as current smokers.

4.2 Visualisations of the Analytical Sample

To further enhance the knowledge and understanding of the final analytical sample, the distributions of both the primary predictor (ALMI), the continuous form of the outcome (HbA1c), and the relationship between the can be seen below.

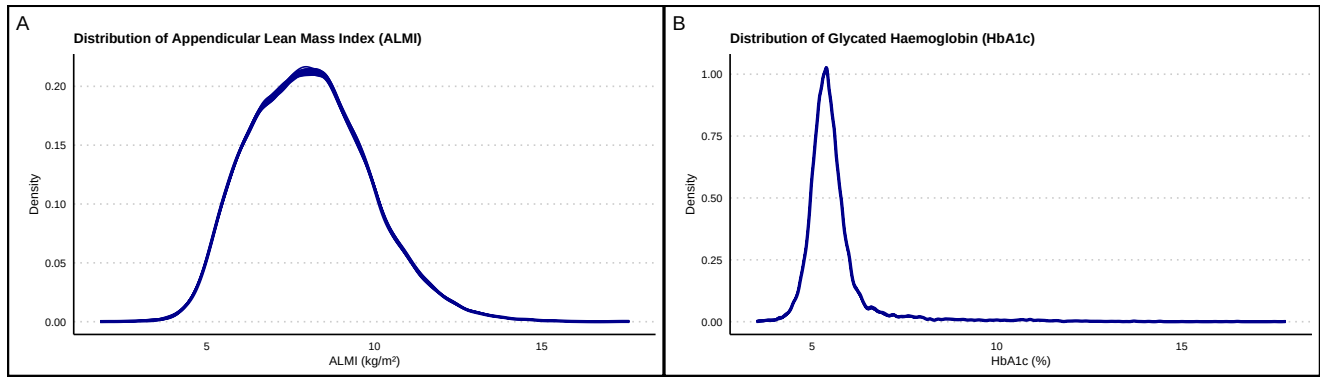


Figure 1: A: Density distribution of Appendicular Lean Mass Index (ALMI) in the final analytical sample. Each blue line corresponds to the distribution of ALMI in each of the 30 imputed datasets. B: Density distribution of Glycated Haemoglobin (HbA1c) in the final analytical sample, also over each imputed dataset.

Figure 1 shows the distribution of ALMI and HbA1c in each imputed dataset. The ALMI distribution is approximately normal and centred around 8.5 kg/m². There is very little variation in the distributions of different imputations, highlighting the low uncertainty introduced through the imputation process. The HbA1c distribution appears to be uni-modal and highly right-skewed, with the majority of the population clustered around a healthy range (around 5.2%). The very little variation across imputations is also evident here.

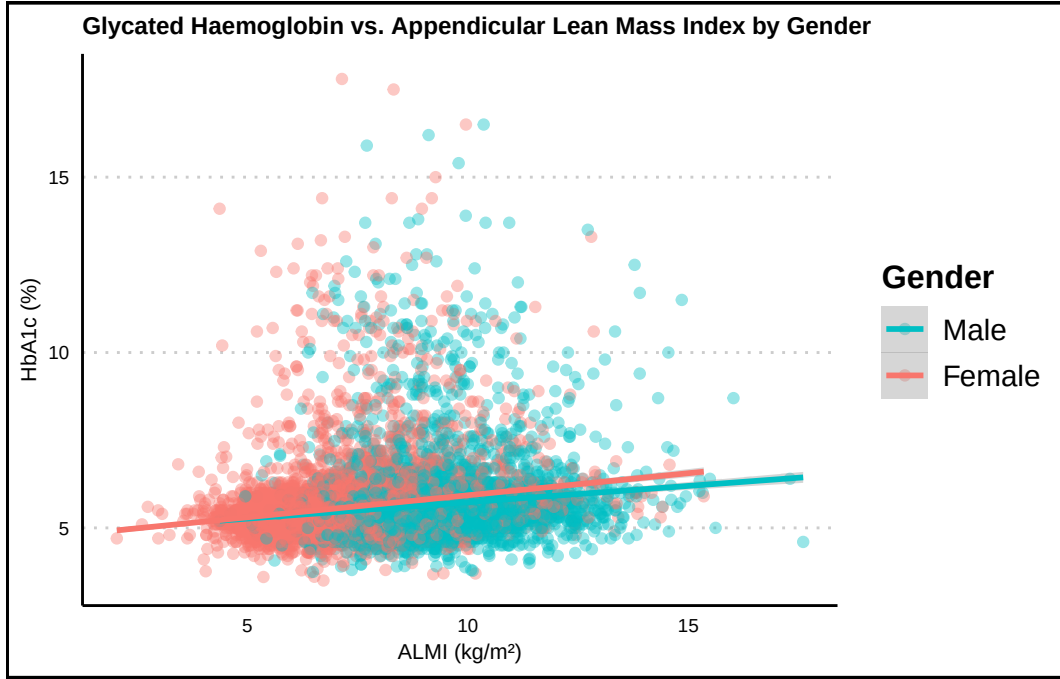


Figure 2: Scatter plot showing the relationship between ALMI and HbA1c in imputation 1 of the final analytical sample, stratified by gender. Points represent individual participants in the first imputed dataset. Solid lines represent the linear model trend line for each gender.

Finally, Figure 2 illustrates the unadjusted relationship between ALMI and HbA1c in the first imputed dataset. The majority of points cluster around 5% HbA1c; however, many outliers exhibit very high HbA1c levels across a range of ALMI values, well above the 6.5% diabetic threshold. These are the cause of the right-skewed density plot of HbA1c in Figure 1. It is clear that on average, men have a higher ALMI than women, with most of the blue points clustered to the right of the red points. For both men and women, higher ALMI appears to be associated with higher HbA1c. This counter-intuitive observation will later be demonstrated to be an artefact of confounding.

4.3 Hierarchical Model Building and Key Findings

The methodological and hierarchical model-building process, used to understand the influence of various confounding variables and choose robust final models for both the Onset and Progression outcomes, is summarised in Table 2 and Table 3, respectively. These tables track the change in the ALMI coefficient, its odds ratio, and key model diagnostics as logical blocks of, and individual variables were sequentially added, allowing for a comprehensive understanding of why the final models were chosen. Each model is the result of pooling across all 30 imputations as outlined in Section 3.5.2.

The stepwise model-building of the Onset models summarised in Table 2 illustrates how before

Table 2: Hierarchical Adjustment of the Association Between ALMI and the Onset of Glycaemic Dysregulation.

Odds Ratios, Coefficients, Percentage Changes, and Diagnostics							
Model	Covariate Changes	ALMI Odds Ratio (OR)	95% Confidence Interval (OR)	ALMI Coefficient (log OR)	Percentage Change in Coefficient	McFadden Pseudo R^2	Archer- Lemeshow GoF Test P-value
Model 1	Crude	1.289	(1.239, 1.342)	0.254	N/A	0.032	0.264
Model 2	+ Demographics	1.468	(1.397, 1.542)	0.384	51.06%	0.164	0.734
Model 3	+ Body Fat Percentage (%)	1.350	(1.273, 1.432)	0.300	-21.79%	0.172	0.996
Model 4	+ Waist Circumference (cm)	0.997	(0.919, 1.081)	-0.003	-101.07%	0.186	0.922
Model 5	+ Poverty Income Ratio	1.019	(0.938, 1.106)	0.019	681.01%	0.190	0.943
Model 6	+ Physical Activity (mins)	1.020	(0.939, 1.108)	0.020	7.42%	0.190	0.949
Model 7	+ Healthy Eating Index	1.026	(0.945, 1.114)	0.026	30.06%	0.192	0.878
Model 8	+ Alcohol Status	1.023	(0.942, 1.110)	0.022	-14.48%	0.194	0.990
Model 9	+ Smoking Status	1.042	(0.960, 1.131)	0.041	84.81%	0.196	0.987
Model 10	+ Average Daily Sleep (hrs)	1.040	(0.958, 1.129)	0.039	-4.25%	0.196	0.991

adjusting for any confounders in the crude model, the odds ratio of ALMI was 1.29 with a 95% CI of [1.240, 1.342]. This implies that a 1 kg/m² increase in ALMI is associated with an increase in the odds of having elevated HbA1c by 29%. This result is counter-intuitive and comes as a result of a failure to control for confounding. The inclusion of the demographic variables and body fat percentage resulted in significant fluctuations of the ALMI coefficient, where significant is defined as a change larger than 10%; however, it still remained positive and statistically significant, resulting in an odds ratios larger than 1.

In Model 4, once waist circumference was included, the ALMI coefficient became statistically insignificant, with its 95% CI including 0. The addition of further covariates up to the fully adjusted Model 10 did not change this, with ALMI remaining non-significant throughout. The loss of significance after including waist circumference suggests that the initial positive association may have reflected confounding by central adiposity. The earlier models (prior to Model 4) indicate that participants with higher muscle mass also tended to have other characteristics associated with higher risk, specifically larger waist circumference.

The McFadden Pseudo R^2 values can be seen to increase throughout the stepwise process. The addition of the demographic variables resulted in its largest jump from 0.033 to 0.165. The fully adjusted Model 10 had a value of 0.196, meaning that even after comprehensive adjustment of all confounders, it still cannot be thought to have an “excellent” fit, as discussed in Section 3.6. Importantly, the Archer-Lemeshow goodness-of-fit tests were non-significant for all ten models (all $p > 0.05$), confirming that each model in the sequence fit the data well.

The “Percentage Change in Coefficient” column demonstrates the sensitivity of the ALMI coefficient to the sequential addition of confounders. While the coefficient’s estimate continued to change by more than 10% in subsequent models, this criterion is no longer a meaningful guide for selecting a parsimonious model, as the primary association had been fully attenuated. Since the effect of ALMI became statistically insignificant in Model 4 and remained so thereafter, the 10%

rule was not used to select a final model for this outcome.

The primary finding from the Onset modelling process is that ALMI is not significantly associated with glycaemic dysregulation. A final model must show that this result holds with appropriate confounder adjustment; therefore, the fully adjusted Model 10 was chosen in this case. Selecting the fully adjusted model ensures that the null association of ALMI with onset is robust to adjustment for all measured confounders.

Table 3: Hierarchical Adjustment of the Association Between ALMI and the Progression to Diabetes

Odds Ratios, Coefficients, Percentage Changes, and Diagnostics							
Model	Covariate Changes	ALMI Odds Ratio (OR)	95% Confidence Interval (OR)	ALMI Coefficient (log OR)	Percentage Change in Coefficient	McFadden Pseudo R ²	Archer- Lemeshow GoF Test P-value
Model 1	Crude	1.388	(1.315, 1.464)	0.328	N/A	0.044	0.165
Model 2	+ Demographics	1.583	(1.466, 1.709)	0.459	40.13%	0.159	0.390
Model 3	+ Body Fat Percentage (%)	1.462	(1.338, 1.598)	0.380	-17.30%	0.164	0.708
Model 4	+ Waist Circumference (cm)	0.793	(0.691, 0.910)	-0.232	-161.20%	0.216	0.705
Model 5	+ Poverty Income Ratio	0.805	(0.702, 0.923)	-0.217	6.49%	0.217	0.905
Model 6	+ Physical Activity (mins)	0.810	(0.706, 0.929)	-0.210	3.16%	0.219	0.924
Model 7	+ Healthy Eating Index	0.811	(0.707, 0.931)	-0.209	0.65%	0.220	0.948
Model 8	+ Alcohol Status	0.806	(0.702, 0.925)	-0.216	-3.37%	0.223	0.968
Model 9	+ Smoking Status	0.803	(0.697, 0.924)	-0.220	-1.64%	0.223	0.897
Model 10	+ Average Daily Sleep (hrs)	0.803	(0.697, 0.924)	-0.220	-0.14%	0.223	0.867

The immediate difference between the results of the Onset and Progression outcomes is that in the Progression setting, seen in Table 3, the ALMI coefficient is statistically significant for every iteration of the model-building process. This implies that ALMI is associated with the transition to clinical diabetes; therefore, the goal was to select a final model that adjusts for confounding while avoiding unnecessary complexity. This will be done using the 10% rule. The final model can then be used to make a conclusive statement regarding the association of skeletal muscle with the odds of becoming diabetic.

The paradoxical positive association seen in the crude Onset model (Model 1) also translates to the crude Progression model. Here, ALMI is associated with a 39% increase in the odds of becoming diabetic, with a 95% CI of [1.319, 1.463]. As also seen with the Onset models, this counter-intuitive finding persists after adjusting for key demographics and body fat percentage.

Where the Progression models differ from the Onset models comes after the inclusion of waist circumference in Model 4. Where the Onset Model 4 saw the association for ALMI lose statistical significance, the Progression Model 4 results in its dramatic inversion to a statistically significant and protective odds ratio of 0.788, with a 95% CI of [0.683, 0.910]. This represents a -162.6% change in the coefficient, providing clear evidence of the strong confounding effect of waist circumference that was masking the true protective association of muscle mass in the simpler models.

As further lifestyle and socio-economic confounders were added in Models 5 through 10, the odds ratio for ALMI remained stable and statistically significant, confirming the robustness of this

protective association. After Model 4, the inclusion of further predictors did not change the coefficient for ALMI by more than 6%. The final, fully adjusted model (Model 10) yielded an odds ratio of 0.796.

The McFadden’s Pseudo R^2 values increased in a similar manner as they did throughout the Onset models, with the largest increase being from 0.044 to 0.157 from Model 1 to Model 2. However, from Model 4 onwards, they reached values of above 0.2, indicating an “excellent” fit. The Archer-Lemeshow goodness-of-fit tests were non-significant for all 10 models ($p > 0.05$), meaning that every model was well-defined.

The final Progression model from which conclusions can be drawn from is Model 4. This is due to the ALMI coefficient changing by no more than 6% after the inclusion of any further covariates, justifying the choice of Model 4 on the basis of parsimony. This model contains only the predictors that are significant confounders of the association of muscle and diabetic status. The odds ratio of ALMI in Model 4 can be used to infer that an increase in ALMI by 1 kg/m² is associated with a 21.2% reduction in the odds of progressing to diabetic levels of blood glucose.

In addition, survey-weighted variance inflation factors (VIFs) and further exploratory modelling provided a clear statistical explanation for the observed coefficient changes. In Model 3 for both outcomes, which did not include waist circumference, all VIF values were below 2.5, demonstrating no significant multicollinearity. However, upon the inclusion of waist circumference in Model 4, the VIF values for the three body composition variables rose sharply, confirming a high degree of multicollinearity (VIFs of 4.5, 6.5, and 7 for the Onset model; 6, 6, and 9 for the Progression model).

This multicollinearity had different consequences for each outcome. For the Onset outcome, the null effect of ALMI was revealed as soon as waist circumference was added regardless of whether body fat percentage was included, suggesting that central adiposity is the primary confounder of the initial paradoxical association.

In contrast, for the Progression outcome, a more complex suppressor effect was evident. The protective effect of ALMI only became significant when both waist circumference *and* body fat percentage were included in the model, simultaneously. This underscores the critical importance of a comprehensive, two-component body composition adjustment to correctly specify the relationship between muscle mass and the progression to clinical diabetes.

It is also noteworthy that of the covariates included, body fat percentage and waist circumference were strong and statistically significant predictors across both the Onset and Progression models.

The forest plot seen in Figure 3 concisely visualises the key stage-specific findings by plotting the ALMI odds ratio and 95% confidence intervals estimated by Onset Model 10 and Progression Model 4.

``height`` was translated to ``width``.

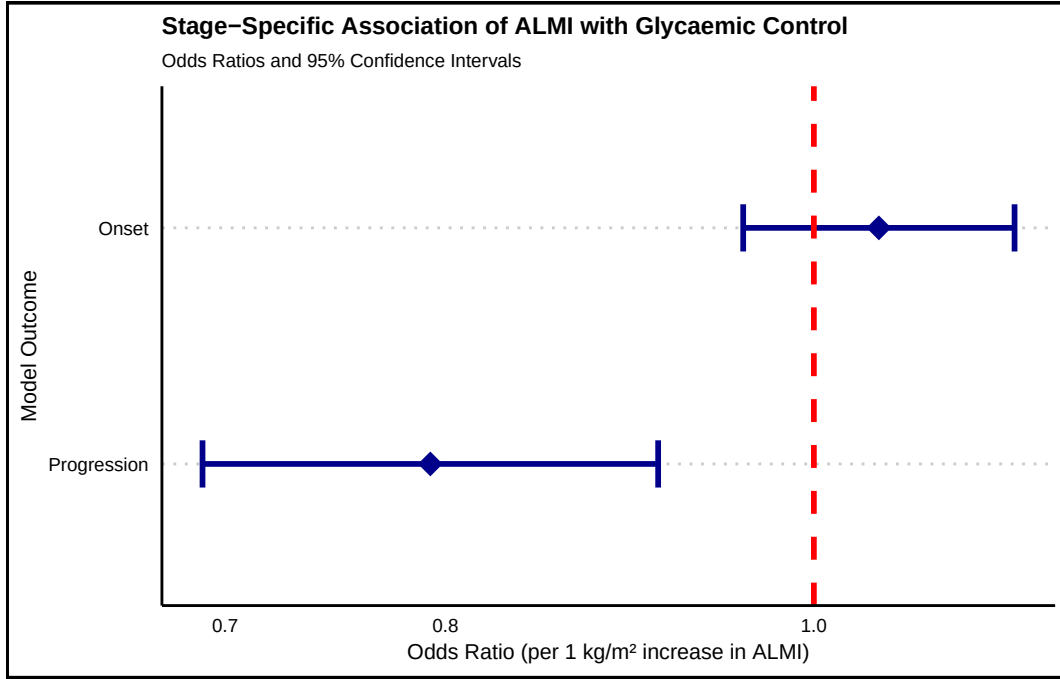


Figure 3: Stage-specific associations of skeletal muscle, proxied by ALMI, with glycaemic control. The plot displays the odds ratio of ALMI estimated by the final Onset and Progression models (10 and 4, respectively), along with their 95% confidence intervals. The dashed red line at 1.0 represents the null association.

The interception of the Onset model line with the dashed red line represents the inclusion of an odds ratio of 1 in the 95% confidence interval. This shows how there is not sufficient evidence to reject the null hypothesis that there is no association between skeletal muscle with the raising of blood glucose levels beyond the normal range.

Conversely, the Progression model line can be seen to the left of the red line, demonstrating the statistically significant protective association of skeletal muscle against the progression to diabetes.

The X-axis in Figure 3 uses a logarithmic scale. This allows for the odds ratios to be appropriately represented. A standard linear scale would misrepresent the symmetrical aspect of the confidence intervals and multiplicative scaling of the odds ratios.

4.4 Final Model Validation and Robustness Checks

Residual diagnostics and influential point analyses are included to confirm the validity of the final models and ensure the robustness of their findings. All of the binned residual plots referenced in this section can be seen in 7.7.1.

The binned residual plot in Figure 6 show no significant patterns, with the majority of the residuals falling within the grey ± 2 standard error boundaries. This indicates that the final Onset model

was well-calibrated. In addition, there were no clear differences between the residuals of each imputation, meaning that the imputation strategy was robust.

The component-wise binned residual plots for Onset Model 10 in Figure 7 evidence that the assumptions of linear covariates in the model were justified. There are no clear curves in the residuals that would suggest higher order terms of covariates would provide significant advantages. The pattern of large residuals in the physical activity plot is likely a product of data sparsity at the extreme end of the distribution, and is not a cause for concern. Furthermore, there is fanning of the residuals, particularly in the plot for age; however, this is also not a concern, as the design-based standard error estimation employed by `svyglm` appropriately handles the potential overdispersion that these plots highlight. Only imputation 1 was used for these plots due to the previously discussed strong evidence of stability across imputations.

In contrast, the residuals for the final Progression model, seen in Figure 8, seem to show a systematic pattern. For very low expected probability of progression to diabetes, there does not appear to be a clear pattern; however, the model seems to systematically underestimate the probability for individuals of low-to-moderate risk and overestimate it for those at very high risk. This provides evidence that the model may be miscalibrated, and that its use for specific probability estimation is not advised. The implications of this finding for the model’s explanatory power will be addressed in the Section 5. Despite this, the residuals stay mostly within the grey ± 2 standard error boundaries, although this does not alleviate the calibration concerns. Similarly to the onset model, the residuals remain consistent across imputations, providing further evidence that the imputation process was robust.

Finally, the component-wise binned residual plots for Progression Model 4 in Figure 9 suggest that the assumption of linear covariates is valid, with no systematic patterns visible in any plot. As was the case for the Onset component wise residual plots, these were formed using only imputation 1 due to the consistency over imputations.

In addition to residual diagnostics, the influential points analysis are as follows: Both Onset Model 10 and Progression Model 4 were refit using reduced datasets that did not contain the observations that were in the top 1% of Cook’s Distance of their original fits. The ALMI coefficient was still statistically insignificant in Onset Model 10, and in Progression Model 4, it was statistically significant and equal to -0.235, corresponding to an odds ratio of 0.79. This meant that the finding of both models are robust and are not driven by extreme outliers in the data.

4.5 Visualising the Association between ALMI and Diabetes

To visualise the shape and magnitude of the association of skeletal muscle mass and diabetes, Progression Model 4 was used to generate probability predictions of becoming diabetic for a range of ALMI values, while holding age, race, and body composition constant at their average across all imputations. This was done for both males and females, as discussed in 3.7. The results of the model predictions are seen in Figure 4.

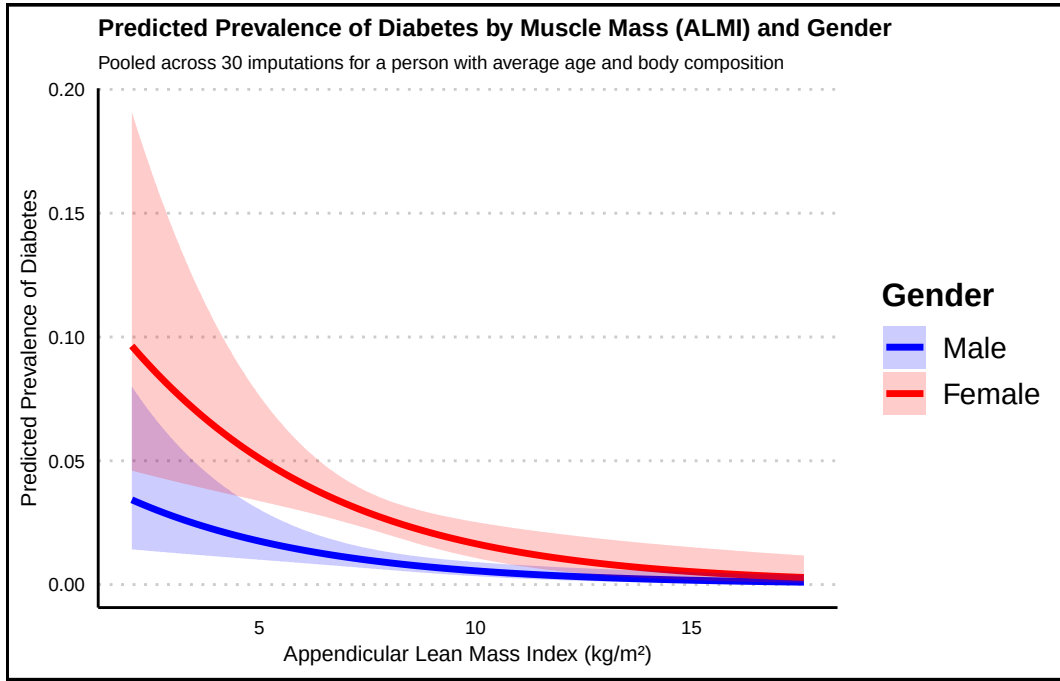


Figure 4: Predicted probability of diabetes across the range of Appendicular Lean Mass Index (ALMI), stratified by gender. Predictions are generated from Progression Model 4, holding all other covariates constant at their mean or mode. Solid lines represent the pooled mean prediction, and the shaded areas represent the corresponding 95% confidence intervals.

This diagram clearly illustrates the strong protective association between skeletal muscle against the likelihood of becoming diabetic, with higher values of ALMI corresponding to lower predicted probabilities. The trend for both men and women is non-linear, with both curves demonstrating a shape similar to exponential decay. Women appear to have a higher probability of diabetes for all amounts of skeletal muscle, with the red mean prediction line always above the blue. However, for ALMI below approximately 5 kg/m² and above 10 kg/m², there is overlap of the confidence intervals between both genders, implying the probability differences between genders is less certain here. The red curve being steeper than the blue also suggests that the protective effects of skeletal muscle are perhaps stronger in women.

The predicted probability at the lowest level of ALMI present in the data (2.1 kg/m²) was 0.1 for females and 0.04 for males; however, predictions made at lower levels of ALMI carry a high degree of uncertainty, highlighted by the large confidence intervals at these levels. This uncertainty is likely due to the lack of data in this range of ALMI. There were only 5 individuals with ALMI below 5 kg/m² in the first imputed dataset. This is assumed to not be significantly different across all other imputations.

Conversely, for ALMI above around 5 kg/m², the confidence intervals are much narrower, meaning predictions here are much more precise. Across all imputed datasets, the mean value of ALMI for females was 7.2 kg/m², and was 9.0 kg/m² for men. The predicted probability of becoming

diabetic for females at their mean ALMI was 0.03, and for men it was 0.007.

While the data was also sparse for large ALMI (only 9 individuals had more than 15 kg/m²), the confidence intervals in this range maintain their restricted width. This is due to a mathematical property of probability. As the model’s predicted probability of diabetes approaches its absolute floor of zero, the potential for variance is naturally compressed. Therefore, the confidence interval is forced to be narrow, as the lower bound is anchored near zero, and the upper bound is unable to significantly vary due to the already low point estimate.

4.6 Cross-Validation Performance

This final subsection of the results will evaluate the findings of the cross-validation approach outlined in Section 3.8 to determine Progression Model 4’s generalisability and clinical usability for classification. This process was not carried out for Onset Model 10 due to the insignificance of ALMI in the model. These findings are summarised in Table 4.

Table 4: Results of Survey Weighted 5-fold Cross-Validation Repeated 10 Times

Metric	Value
AUC	0.840
Optimal Threshold	0.059
Sensitivity	79.78%
Specificity	73.15%
Accuracy	73.57%

Specifically, the table contains the mean of the mean performance metrics over each of the 10 repeats, as a way to concisely summarise the overall performance of the model. The most important statistic, the mean weighted AUC of 0.844, suggests that the model has high discriminatory power.

The mean optimal threshold of 0.061, although not a measure of model performance, is useful to be aware of. This means that on average, an observation was classified as diabetic if they were predicted a probability of either 0.061 or higher of being so. This threshold provided the optimal balance between sensitivity and specificity, resulting in the respective means of 79.1% and 74.8%. The mean overall weighted accuracy was 75.1%.

To aid in the interpretation of the results, the ROC curve in Figure 5 provides a visual representation of the model’s discriminatory ability.

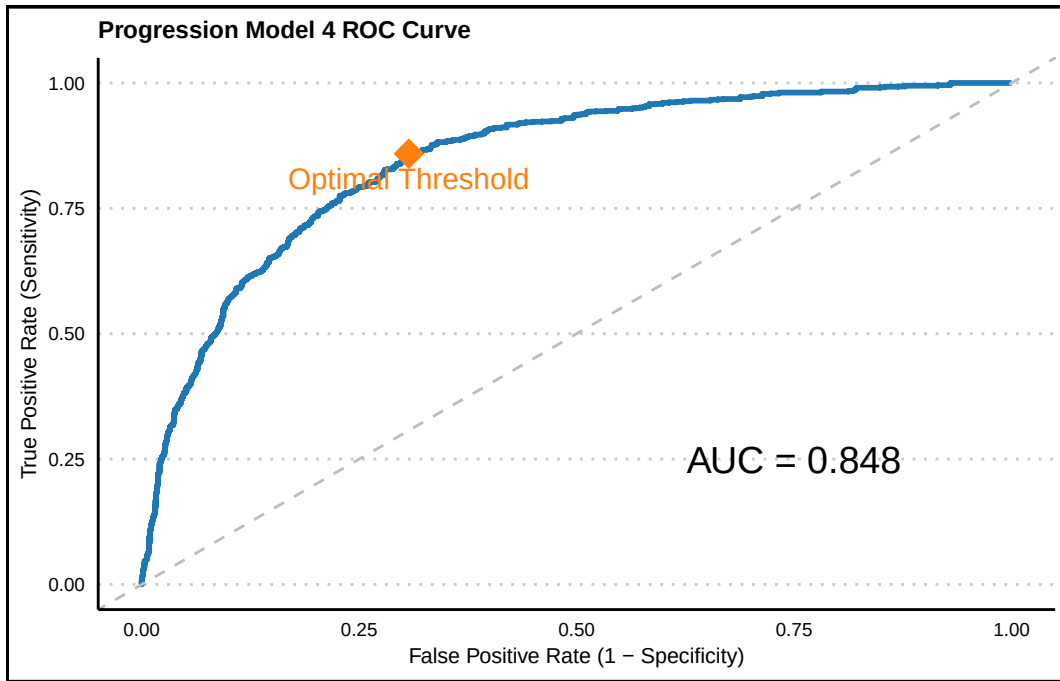


Figure 5: Receiver Operating Characteristic (ROC) curve from the final iteration of the cross-validation for Progression Model 4. The orange diamond highlights the optimal threshold selected by maximising Youden's J statistic. The Area Under the Curve (AUC) of 0.85 indicates high discriminatory power.

The steep initial climb in the ROC curve represents how a high sensitivity is achieved with minimal sacrifice to specificity. The curve sits substantially higher than the grey dotted diagonal, indicating the model's significantly better discriminatory ability than random assignment. The optimal classification threshold found by maximising Youden's J statistic can be seen marked with the orange diamond.

Chapter 5

Discussion

5.1 Summary of Principal Findings

This study, based on the analysis of a large, nationally representative US sample from NHANES, yielded two key findings. The first was that the association between skeletal muscle and glycaemic control appears to be stage-specific. After comprehensive adjustment for confounding, higher skeletal muscle mass was significantly linked to lower odds of progression to clinical diabetes. In other words, among individuals with blood glucose levels in the healthy or prediabetic ranges, greater muscle mass corresponded with reduced odds of developing diabetes. However, no meaningful link was observed between skeletal muscle and the initial onset of dysglycaemia, indicating that higher muscle mass does not relate to a lower likelihood of moving beyond normal glucose levels.

The second key finding was methodological, highlighting a complex interplay between body composition variables that shapes the observed relationships between skeletal muscle and glycaemic regulation. For the onset of dysglycaemia, the initial counter-intuitive positive association between muscle mass and HbA1c was largely attributable to confounding by central adiposity (waist circumference). Once this was accounted for, the underlying null association became evident. Conversely, the apparent protective link between muscle mass and progression to clinical diabetes was revealed only after controlling for both central *and* overall adiposity (body fat percentage).

5.2 Interpretation of Findings in the Context of Existing Literature

The critical stage-specific characteristic of these findings allows for more nuanced inferences into the underlying processes that are associated with poor glycaemic control and the incidence of diabetes. The absence of an independent association between skeletal muscle and the onset of elevated HbA1c indicates that, during the early stages of glycaemic dysregulation, excess adiposity, particularly central adiposity and visceral fat, may play a more dominant role in the observed patterns. This interpretation can be made due to the inference of the coefficients of body fat percentage and waist

circumference themselves in the models. In simpler models that did not include waist circumference, the body fat percentage coefficient was positive, indicating that higher levels of overall adiposity were, as expected, associated with higher odds of glycaemic dysregulation. However, upon the inclusion of waist circumference in Models 4 and upwards, its coefficient became negative, implying a protective association.

Although this may seem counter-intuitive, it highlights a nuanced statistical pattern: for individuals with equal visceral fat, a higher overall body fat percentage reflects more subcutaneous rather than central fat. Subcutaneous fat is associated with lower risk of poor glycaemic control than visceral fat. This pattern was observed for both onset and progression outcomes, with visceral fat (waist circumference) showing stronger associations with risk than subcutaneous fat or skeletal muscle during the onset of dysregulation.

The finding of the null association found between skeletal muscle and the onset of dysglycaemia appears to be a novel one. While numerous studies have found a general protective association between muscle mass and glycaemic control, the literature is inconsistent regarding this initial stage, with some studies reporting a risk association and others a protective one. It is plausible that these conflicting results are a direct consequence of the powerful confounding effect of central adiposity, which this analysis was able to isolate and control for, thereby revealing the true null association for skeletal muscle.

These stage-specific results are consistent with extensive literature identifying visceral adiposity as a powerful, independent determinant of insulin resistance and dysglycaemia. Findings from large imaging-based cohort studies, such as the Framingham Heart Study, suggest that independent of total body fat, visceral adipose tissue (VAT) is more strongly associated to insulin resistance and diabetes than subcutaneous adipose tissue (SAT) [17]. The inversion of the coefficient of body fat percentage in this analysis provides statistical evidence directly supporting these findings.

In direct contrast, the significant protective association of skeletal muscle against the progression to clinical diabetes implies that for individuals with already elevated blood sugar in the prediabetic range, the ability for skeletal muscle to act as metabolic sink for glucose disposal becomes crucial, regardless of the remaining strong association of visceral fat mass and risk. Higher degrees of muscle mass provide a greater capacity to clear glucose from the bloodstream, which may aid in preventing the pancreatic beta-cell dysfunction that leads to the final transition to type 2 diabetes [36].

This finding provides population-level evidence that supports the mechanisms observed in clinical intervention studies. For example, a randomised controlled trial conducted by Luo, X., Wang, Z., Li, B. et al in 2023, found that in a prediabetic cohort, individuals that conducted resistance training (which directly increases muscle mass) over a period of 12 weeks saw significant improvements in glycaemic control and insulin resistance [40]. Furthermore, another group of participants carried out aerobic exercise regiments for the 12 weeks. It was found that aerobic exercise led to decreases in blood glucose levels as a result of weight loss; whereas, the effect of resistance training aided in glycaemic control through other pathways, and was even found to have a stronger protective effect than weight loss.

5.3 Strengths and Limitations

To provide a thorough evaluation of these findings, it is important to highlight both the key strengths and potential limitations of the analysis. Key aspects of the data and methodological rigour supported the quality of the results. The large, survey-weighted NHANES sample provides nationally representative estimates, increasing reliability and generalisability. Additionally, body composition was measured using dual-energy X-ray absorptiometry (DXA), which offers greater accuracy and objectivity than alternative methods such as BIA [43] [56].

In addition to features of the NHANES sample itself, the sophisticated analytical processes used to leverage it were key. Surrounding literature was utilised to inform the choice of the appropriate outcome, predictor, and confounding variables, some of which underwent meticulous harmonisation due to changes in the NHANES survey structure across each cycle. This meant that each variable included in the modelling procedure was meaningful and contributed to the goal of finding final, well-defined models for each outcome variable.

Further data processing, such as the handling of structural missingness as a result of survey skip patterns, was paramount. This made certain that no data was assumed missing when it was not, greatly improving the integrity of the sample. Following the completion of the final analytical sample, the robust multiple imputation strategy was a key strength. This ensured that the advantages of the large sample size were not lost due to the loss of a small proportion of data. Using physiologically informed bounds also guaranteed that all imputed data was realistic, further improving data integrity.

The iterative modelling journey allowed for an intricate sensitivity analysis that resulted in the choice of the final models based on the significance of the primary predictor variable (ALMI), and how its coefficient changed when each confounding variable was included. Each model underwent goodness-of-fit diagnostics using the Archer-Lemeshow goodness-of-fit test for survey-weighted models to ensure they were well-defined, and their McFadden’s Pseudo R^2 was also calculated to heuristically represent their respective explanatory power. In addition to these general diagnostics, further residual checks and influential point analyses were conducted on the final Onset and Progression models to provide added nuance into the quality of their fits.

The use of survey-weighted logistic regression models proved effective in isolating the stage-specific role of skeletal muscle both at the onset of dysglycaemia and during progression to clinical diabetes. In particular, the final Progression model yielded a clear odds ratio that directly quantified the protective association of greater muscle mass against diabetes incidence. The methodological progression - from ordinal regression, to quasi-Poisson binary classification, and finally to quasi-binomial logistic regression - reflected both statistical rigour and close attention to the structure of the data.

Extending the analysis beyond simple coefficient inference was a key strength. By using the final model to generate predictions of diabetes prevalence, the abstract statistical findings were translated into a clear, visual representation of the dose-response relationship between muscle mass and diabetes risk. This step quantifies the real-world magnitude of the protective association, making

the findings more accessible and clinically relevant. The visualisation of the ALMI coefficient in Onset Model 10 and Progression Model 4 further improved the interpretability of the findings.

Furthermore, the rigorous k-fold cross-validation provided a formal assessment of the generalisability of Progression Model 4. This process confirmed the model’s high discriminatory power (AUC = 0.844) on unseen data, demonstrating that its findings are robust and not simply an artefact of overfitting the training sample. Together, these steps enhance the trustworthiness of the study’s conclusions.

Despite these strengths, it is crucial to interpret the findings of this analysis in the context of its potential limitations. The most significant of these is perhaps the cross-sectional nature of the NHANES data. Each data point corresponds to a separate individual at a specific point in time. A consequence of this data structure is that, while it is well-suited for identifying statistical associations, it does not allow these inferences to be extended to causal conclusions. Therefore, while this study demonstrates a strong protective association between muscle mass and the progression to diabetes, it cannot establish that increased muscle directly reduces risk.

Another limitation is the potential miscalibration of the final Progression model, as indicated by residual diagnostics. The binned residual plot suggested a mild tendency for the model to under-predict diabetes risk at lower predicted probabilities and over-predict at higher predicted probabilities. Consequently, the absolute predicted probabilities presented in Section 4.5 should be interpreted with caution, although the overall shape of the prediction curves is likely reliable.

Importantly, while this calibration issue warrants caution for predictive applications, it is unlikely to invalidate the model’s use for explanatory inference. The distinction between predictive calibration and explanatory inference is crucial: while the model may not be perfectly calibrated for prediction, the odds ratios can remain largely reliable estimates of the direction and strength of association. Component-wise residual diagnostics indicated no need for higher-order covariates, and cross-validated AUC values (>0.8) demonstrate excellent discrimination. Therefore, the central conclusions regarding stage-specific, independent associations remain well-supported by the data despite the mild calibration issue.

Other potential limitations include the potential for residual confounding despite the comprehensive adjustments made in the stepwise model building process. There likely exists other confounding variables that if included, may improve the models and make their findings even more robust. Examples of such may include unmeasured factors such as genetics or specific dietary patterns. The inclusion of further variables, or interactions between current ones, may improve the calibration issues seen from the residuals.

Furthermore, many of the included lifestyle variables such as physical activity, alcohol consumption, and smoking habits are based on self-report, which is subject to recall bias. Unfortunately, this is unavoidable when including such variables; however, it is worth acknowledging. Another unavoidable limitation of using survey data is the inconsistency in questionnaire structure across survey cycles. Some of these inconsistencies were effectively addressed, as described in Section 3.2.4; however, some remained. For example, lifetime drinking status in the 2011–2016 cycles was assumed to be defined by whether an individual had consumed at least 12 alcoholic drinks in

their lifetime, since no question asked about ever having a single drink, unlike in the 2017–2018 cycle. Additionally, the frequency of alcohol consumption in the 2017–2018 cycle was converted from a categorical to a continuous variable to align with the already continuous measure from the 2011–2016 cycles, which inevitably introduced some error due to the assumptions required.

5.4 Implications and Future Directions

The implications of this research in the realm of public health are notable. The specific nuances involved in the mechanisms associated with the onset of dysglycaemia and progression to diabetes imply that a “one-size-fits-all” approach may be too simplistic.

In the early stages of increasing blood sugar, central adiposity possessed the primary independent association with increased risk, with this association also extending to the progression to diabetic HbA1c. Therefore, public health bodies should continue to emphasise the importance of improving diet and increasing general exercise, both of which contribute to reducing visceral fat storages. However, for individuals already categorised as pre-diabetic, added emphasis should be placed upon encouraging resistance training to maintain and/or increase muscle mass, due to this analysis showing that higher levels of skeletal muscle are associated with a reduction in the odds of progression to diabetes from a pre-diabetic state.

From the lens of conducting further research on this topic, the results of this study highlight the importance of comprehensive body composition assessment. The inclusion of muscle mass, central fat, and overall fat in the statistical models is crucial to minimise the risk of obscuring or misinterpreting the true independent associations of each component.

Future studies should extend the findings in this paper by investigating causality. The primary limitation of this study was the cross-sectional design of the data. Exploration of the documented associations in a longitudinal cohort study would have great clinical utility, and would provide stronger evidence that having greater muscle mass precedes a reduction in the risk of progressing from pre-diabetes to diabetes.

Furthermore, if researchers seek to adapt Progression Model 4 as a clinical tool for risk stratification, due to potential calibration issue highlighted by its binned residual plot, recalibration techniques such as logistic recalibration or isotonic regression could be applied post-estimation to improve the accuracy of absolute risk predictions. Additionally, although component-wise residual diagnostics suggested that higher order covariates were not required, it may prove beneficial to explore more flexible modelling of the continuous predictors by using methods such as splines, as this may reduce subtle miscalibration.

It would also be advantageous to explore how the findings evolve with the inclusion of additional, more detailed potential confounding variables - such as muscle quality in addition to muscle quantity, or genetic risk scores for diabetes - to examine potential interactions with body composition. Specific covariate interaction terms could also be included, as these may further improve the models.

Chapter 6

Conclusion

This dissertation sought to discover the independent and stage-specific role of skeletal muscle in glycaemic control. After a comprehensive analysis, it can be concluded that this role is indeed stage-specific. Higher skeletal muscle mass has a protective association against the progression to clinical diabetes. Specifically, it was found that a $1\text{kg}/\text{m}^2$ increase in appendicular lean mass index was associated with approximately a 21.2% reduction in the odds of becoming diabetic. Conversely, it had no independent association with the initial onset of glycaemic dysregulation.

The research began with the preliminary hypothesis that increases in skeletal muscle would be protective across all stages of glycaemic control. However, the initial results were unexpectedly illogical, suggesting that muscle mass possessed a harmful association across all stages. This prompted a more detailed investigation, which ultimately uncovered the true, nuanced, stage-specific relationship. When modelling for the onset of dysglycaemia, the inclusion of waist circumference as a confounding variable, and proxy for central adiposity, revealed the true null association of muscle mass. In contrast, when modelling the progression to diabetes, a two-component adjustment for both central adiposity and overall adiposity, proxied by body fat percentage, uncovered the aforementioned protective association. A comprehensive sensitivity analysis including adjustment for confounding variables of demographics and lifestyle factors as well as body composition, confirmed that the findings were robust and statistically valid. This process highlighted that the true nature of the associations are more intricate than initially hypothesised.

Several recommendations for future research can be made on the basis of these findings. To extend the association results to inferences of causality, the analysis of longitudinal data is highly important. This would instill further confidence that skeletal muscle and glycaemic control are not spuriously correlated. All future modelling approaches should be sure to control for both central and overall adiposity so that their complex confounding relationship can be handled appropriately and results are not obscured. Furthermore, the addition of potential confounding variables not included in this analysis would allow for added insights into the underlying mechanisms, possibly leading to significant coefficient changes and more sophisticated models. The inclusion of higher orders of the included continuous covariates, or interaction terms, may also add statistically significant nuance to the findings. Finally, if the prediction of direct risk probabilities using the

Progression models outlined in this analysis is relevant, recalibration techniques should be employed to improve their accuracy due to residual diagnostics suggesting slight miscalibration for this purpose, although their use for explanatory inference remains valid.

This dissertation makes two key contributions to the fields of metabolic epidemiology and data science in health research. Clinically, it provides novel population-level evidence that the role of skeletal muscle in the control and regulation of blood glucose levels is stage-specific, quantifying this role at each stage. Methodologically, it highlights the critical importance of comprehensively adjusting for body composition in the statistical models, and that failure to do so may lead to misinterpretation of the true, independent role of skeletal muscle.

Chapter 7

Appendices

The following appendices contain all the R code needed to reproduce the results of this research paper. Due to inherent randomness in some of the included procedures, rerunning this code may yield marginally different numerical outputs to those outlined in this report. These differences *do not* affect the interpretation or conclusions drawn in this work.

7.1 Appendix A: Loading packages

```
# ----- #
# Loading required packages for the analysis
# ----- #

library(nhanesA) # For NHANES data
library(survey) # For incorporation of survey design elements
library(systemfonts) # For fonts
library(showtext) # Also for fonts
library(Amelia) # For multiple imputation
library(parallel) # For parallel processing
library(mice) # For more imputation tools
library(mitools) # For even MORE imputation tools
library(svydiags) # For survey model diagnostics
library(poliscidata) # For fit.svyglm function
library(tidyverse) # For tidyverse functions
library(heiscore) # For HEI scores
library(car) # For useful analysis tools
library(svydiags) # For diagnostics of svyglms
library(miceadds) # For micombine.F
library(kableExtra) # For tables
library(arm) # For binned residual plots
library(gridExtra) # For presentation of visualisations
library(WeightedROC) # For survey weighted ROC and AUC calculations
```

7.2 Appendix B: Loading NHANES data

```
# ----- #
# Loading the necessary data from NHANES
# ----- #
```

```

# 2011-12

# Demographics
demo1112 <- nhanes("DEMO_G")
# Body Measures (for height measurements to calculate ALMI)
body1112 <- nhanes("BMX_G")
# Glycohemoglobin (HbA1c)
HbA1c1112 <- nhanes("GHB_G")
# DXA data (for appendicular lean mass)
dxa1112 <- nhanes("DXX_G")
# Physical Activity
phys1112 <- nhanes("PAQ_G")
# Alcohol consumption
alcohol1112 <- nhanes("ALQ_G")
# Cigarette smoking data
smoking1112 <- nhanes("SMQ_G")
# Healthy Eating Score
hei1112 <- score(method = "simple", years = "1112", component = "total score")
# Sleep data
sleep1112 <- nhanes("SLQ_G")

# 2013-14

demo1314 <- nhanes("DEMO_H")
body1314 <- nhanes("BMX_H")
HbA1c1314 <- nhanes("GHB_H")
dxa1314 <- nhanes("DXX_H")
phys1314 <- nhanes("PAQ_H")
alcohol1314 <- nhanes("ALQ_H")
smoking1314 <- nhanes("SMQ_H")
hei1314 <- score(method = "simple", years = "1314", component = "total score")
sleep1314 <- nhanes("SLQ_H")

# 2015-16

demo1516 <- nhanes("DEMO_I")
body1516 <- nhanes("BMX_I")
HbA1c1516 <- nhanes("GHB_I")
dxa1516 <- nhanes("DXX_I")
phys1516 <- nhanes("PAQ_I")
alcohol1516 <- nhanes("ALQ_I")
smoking1516 <- nhanes("SMQ_I")
hei1516 <- score(method = "simple", years = "1516", component = "total score")
sleep1516 <- nhanes("SLQ_I")

# 2017-18

# Demographics
demo1718 <- nhanes("DEMO_J")
body1718 <- nhanes("BMX_J")
HbA1c1718 <- nhanes("GHB_J")
dxa1718 <- nhanes("DXX_J")
phys1718 <- nhanes("PAQ_J")
alcohol1718 <- nhanes("ALQ_J")
smoking1718 <- nhanes("SMQ_J")
hei1718 <- score(method = "simple", years = "1718", component = "total score")
sleep1718 <- nhanes("SLQ_J")

```

7.3 Appendix C: Data processing

```

# ----- #
# Combining the various survey cycles and carrying feature engineering and variable harmonisation

```

```

# ----- #

# Helper function for inner join of multiple datasets
multi_inner_join <- function(dfs) {
  reduce(dfs, ~inner_join(.x, .y, by = "SEQN"))
}

# Create lists of datasets for each time period
dfs1112 <- list(demo1112, body1112, HbA1c1112, dexa1112, phys1112, alcohol1112,
  smoking1112, hei1112, sleep1112)

dfs1314 <- list(demo1314, body1314, HbA1c1314, dexa1314, phys1314, alcohol1314,
  smoking1314, hei1314, sleep1314)

dfs1516 <- list(demo1516, body1516, HbA1c1516, dexa1516, phys1516, alcohol1516,
  smoking1516, hei1516, sleep1516)

dfs1718 <- list(demo1718, body1718, HbA1c1718, dexa1718, phys1718, alcohol1718,
  smoking1718, hei1718, sleep1718)

# Applying function and adding variable denoting time period to aid with variable harmonisation
xsec.data1112 <- multi_inner_join(dfs1112) %>%
  mutate(SurveyCycle = "2011-2012")
xsec.data1314 <- multi_inner_join(dfs1314) %>%
  mutate(SurveyCycle = "2013-2014")
xsec.data1516 <- multi_inner_join(dfs1516) %>%
  mutate(SurveyCycle = "2015-2016")
xsec.data1718 <- multi_inner_join(dfs1718) %>%
  mutate(SurveyCycle = "2017-2018")

# Array of variables needed from each time period
vars1112 <- c(
  "SEQN", "SDMVPSU", "SDMVSTRA", "WTMEC2YR", "RIAGENDR", "RIDAGEYR", "RIDRETH3", "INDFMPIR", # Survey structure and
  # demographics
  "BMXHT", "BMXWAIST", # Body measures
  "LBXGH", # HbA1c
  "DXDRALE", "DXDLALE", "DXDRLE", "DXDLLE", "DXDTPF", # Lean mass in the limbs excl. bone mass
  "PAQ665", "PAD675", "PAQ650", "PAD660", # Physical activity
  "ALQ120Q", "ALQ130", "ALQ101", "ALQ110", # Alcohol consumption
  "SMQ040", "SMQ020", # Smoking habits
  "score", # Healthy Eating Index score
  "SLD010H", # Sleep
  "SurveyCycle" # Marker for survey years
)

vars1314 <- c(
  "SEQN", "SDMVPSU", "SDMVSTRA", "WTMEC2YR", "RIAGENDR", "RIDAGEYR", "RIDRETH3", "INDFMPIR",
  "BMXHT", "BMXWAIST",
  "LBXGH",
  "DXDRALE", "DXDLALE", "DXDRLE", "DXDLLE", "DXDTPF",
  "PAQ665", "PAD675", "PAQ650", "PAD660",
  "ALQ120Q", "ALQ130", "ALQ101", "ALQ110",
  "SMQ040", "SMQ020",
  "score",
  "SLD010H",
  "SurveyCycle"
)

vars1516 <- c(
  "SEQN", "SDMVPSU", "SDMVSTRA", "WTMEC2YR", "RIAGENDR", "RIDAGEYR", "RIDRETH3", "INDFMPIR",
  "BMXHT", "BMXWAIST",
  "LBXGH",
  "DXDRALE", "DXDLALE", "DXDRLE", "DXDLLE", "DXDTPF",
  "PAQ665", "PAD675", "PAQ650", "PAD660",

```

```

"ALQ120Q", "ALQ130", "ALQ101", "ALQ110",
"SMQ040", "SMQ020",
"score",
"SLD012",
"SurveyCycle"
)

vars1718 <- c(
  "SEQN", "SDMVPSU", "SDMVSTRA", "WTMEC2YR", "RIAGENDR", "RIDAGEYR", "RIDRETH3", "INDFMPPIR",
  "BMXHT", "BMXWAIST",
  "LBXGH",
  "DXDRALE", "DXDLALE", "DXDRLE", "DXDLLE", "DXDTPF",
  "PAQ665", "PAD675", "PAQ650", "PAD660",
  "ALQ121", "ALQ130", "ALQ111",
  "SMQ040", "SMQ020",
  "score",
  "SLD012", "SLD013",
  "SurveyCycle"
)

# Extracting variables for each time period
imputation_vars1112 <- xsec.data1112 %>% select(all_of(vars1112))
imputation_vars1314 <- xsec.data1314 %>% select(all_of(vars1314))
imputation_vars1516 <- xsec.data1516 %>% select(all_of(vars1516))
imputation_vars1718 <- xsec.data1718 %>% select(all_of(vars1718))

# Combining
imputation_vars_full <- bind_rows(
  imputation_vars1112,
  imputation_vars1314,
  imputation_vars1516,
  imputation_vars1718
)

# Need to set PAD675 to 0 where PAQ665 is "No" AND set PAD660 to 0 where PAQ650 is "No".
# Also need to set PAD675 and PAD660 to NA whenever they are 9999

# Proportion missing of minutes of moderate and vigorous physical activity
sum(is.na(imputation_vars_full$PAD660))/nrow(imputation_vars_full) # 69% missing
sum(is.na(imputation_vars_full$PAD675))/nrow(imputation_vars_full) # 56% missing

imputation_vars_full <- imputation_vars_full %>%
  mutate(
    # For Vigorous Activity (PAD660)
    # If PAQ650 is 2 ('No') AND PAD660 is currently NA, change it to 0.
    # Otherwise, leave it as it is.
    PAD660 = ifelse(PAQ650 == "No" & is.na(PAD660), 0, PAD660),

    # Assigning NA for "Don't know" code
    PAD660 = ifelse(PAD660 == 9999, NA_real_, PAD660),

    # For Moderate Activity (PAD675)
    # If PAQ665 is 2 ('No') AND PAD675 is currently NA, change it to 0.
    # Otherwise, leave it as it is.
    PAD675 = ifelse(PAQ665 == "No" & is.na(PAD675), 0, PAD675),

    # Assigning NA for "Don't know" code
    PAD675 = ifelse(PAD675 == 9999, NA_real_, PAD675)
  ) %>%
  select(-c(PAQ665, PAQ650))

# Proportion missing of minutes of moderate and vigorous physical activity after replacement
sum(is.na(imputation_vars_full$PAD660))/nrow(imputation_vars_full) # 0.04% missing
sum(is.na(imputation_vars_full$PAD675))/nrow(imputation_vars_full) # 0.09% missing

```

```

# ---- Cleaning alcohol variables and managing sick quitter effect ---- #

imputation_vars_full.2 <- imputation_vars_full %>%

# --- Step 1: Harmonise the two different frequency variables into one ---
mutate(
  Drinking_Days_Per_Year = case_when(

    # For older cycles, use the direct numeric value from ALQ120Q
    SurveyCycle %in% c("2011-2012", "2013-2014", "2015-2016") & ALQ120Q %in% c(777, 999) ~ NA_real_,
    SurveyCycle %in% c("2011-2012", "2013-2014", "2015-2016") ~ ALQ120Q,

    # For the 2017-2018 cycle, convert the categories from ALQ121
    SurveyCycle == "2017-2018" ~ case_when(
      ALQ121 %in% c(77, 99) ~ NA_real_,
      ALQ121 == "Never in the last year" ~ 0,      # Never
      ALQ121 == "Every day" ~ 365,      # Every day
      ALQ121 == "Nearly every day" ~ 312,      # Nearly every day (~6/wk)
      ALQ121 == "3 to 4 times a week" ~ 182,      # 3-4 times a week (~3.5/wk)
      ALQ121 == "2 times a week" ~ 104,      # 2 times a week
      ALQ121 == "Once a week" ~ 52,      # Once a week
      ALQ121 == "2 to 3 times a month" ~ 30,      # 2-3 times a month (~2.5/mo)
      ALQ121 == "Once a month" ~ 12,      # Once a month
      ALQ121 == "7 to 11 times in the last year" ~ 9,      # 7-11 times a year (avg 9)
      ALQ121 == "3 to 6 times in the last year" ~ 4.5,      # 3-6 times a year (avg 4.5)
      ALQ121 == "1 to 2 times in the last year" ~ 1.5,      # 1-2 times a year (avg 1.5)
      TRUE ~ NA_real_
    )
  ) %>%

# --- Step 2: Clean the "drinks per day" variable (ALQ130) ---
mutate(
  # First, handle the special codes for Refused/Don't Know
  Clean_ALQ130 = case_when(
    ALQ130 %in% c(777, 999) ~ NA_real_,
    TRUE ~ ALQ130
  ),

  # Now, use harmonised variable to set non-drinkers to 0
  # handles the structural missingness correctly
  Clean_ALQ130 = ifelse(Drinking_Days_Per_Year == 0, 0, Clean_ALQ130)
) %>%

# --- Step 3: Calculate the final average daily intake ---
mutate(
  AvgDailyDrinks = (Clean_ALQ130 * Drinking_Days_Per_Year) / 365
) %>%

# --- Step 4: Cleaning ALQ110 and creating the single LifetimeDrinkerFlag ---
mutate(
  Clean_ALQ110 = case_when(
    # If they are a current regular drinker (ALQ101 == 1), they must have drunk in their lifetime.
    SurveyCycle %in% c("2011-2012", "2013-2014", "2015-2016") & ALQ101 == "Yes" & is.na(ALQ110) ~ "Yes",
    ALQ110 %in% c(7, 9) ~ NA_character_,
    # Otherwise, keep the original ALQ110 value for now
    SurveyCycle %in% c("2011-2012", "2013-2014", "2015-2016") ~ as.character(ALQ110),
    TRUE ~ NA_character_ # This column doesn't exist in the 2017-2018 cycle
  )
) %>%

mutate(
  LifetimeDrinkerFlag = case_when(
    SurveyCycle %in% c("2011-2012", "2013-2014", "2015-2016") & Clean_ALQ110 == "No" ~ "Never",

```

```

SurveyCycle %in% c("2011-2012", "2013-2014", "2015-2016") & Clean_ALQ110 == "Yes" ~ "Ever",

SurveyCycle == "2017-2018" & ALQ111 == "No" ~ "Never",
SurveyCycle == "2017-2018" & ALQ111 == "Yes" ~ "Ever",

# Handle any remaining special codes
ALQ111 %in% c(7, 9) ~ NA_character_,

TRUE ~ NA_character_
)
) %>%

# --- Step 5: Correct AvgDailyDrinks for Never drinkers ---
mutate(
  AvgDailyDrinks = ifelse(LifetimeDrinkerFlag == "Never", 0, AvgDailyDrinks),

  # Convert the character column to a factor and set the levels
  LifetimeDrinkerFlag = factor(LifetimeDrinkerFlag,
                                levels = c("Never", "Ever"))
) %>%

# --- Step 6: Final selection ---
select(-c("ALQ120Q", "ALQ130", "ALQ101", "ALQ110", "ALQ121", "ALQ111", "Drinking_Days_Per_Year", "Clean_ALQ110",
          "Clean_ALQ130"))

# ---- Cleaning smoking variables ---- #

imputation_vars_full.3 <- imputation_vars_full.2 %>%
mutate(
  Smoking_Status = factor(case_when(

    # Condition 1: If they answered "No" to ever smoking 100 cigarettes, they are a Never Smoker.
    SMQ020 == "No" ~ "Never Smoker",

    # Condition 2: If they HAVE smoked 100 cigarettes (SMQ020 == 1),
    # then look at their answer to SMQ040.
    SMQ020 == "Yes" ~ case_when(
      # If they smoke "Every day" or "Some days", they are a Current Smoker.
      SMQ040 %in% c("Every day", "Some days") ~ "Current Smoker",
      # If they answered "Not at all", they are a Former Smoker.
      SMQ040 == "Not at all" ~ "Former Smoker",
      # Handle any remaining special codes (like Refused/Don't Know) as NA
      TRUE ~ NA_character_
    ),

    # A fallback for any other case (e.g., if SMQ020 itself is missing, refused or don't know)
    TRUE ~ NA_character_
  ),
  # Set the order of the factor levels for clear regression output
  levels = c("Never Smoker", "Former Smoker", "Current Smoker"))
) %>%
select(-c("SMQ020", "SMQ040"))

# ---- Cleaning sleep data ---- #

imputation_vars_full.4 <- imputation_vars_full.3 %>%
mutate(
  # Create a single, harmonised sleep variable in hours
  AvgNightlySleep = case_when(

    # For 2011-2014 cycles, use SLD010H
    SurveyCycle %in% c("2011-2012", "2013-2014") & SLD010H %in% c(77, 99) ~ NA_real_,
    SurveyCycle %in% c("2011-2012", "2013-2014") ~ SLD010H, # Treat the top-code of 12 as 12

```

```

# For the 2015-2016 cycle, use SLD012
SurveyCycle == "2015-2016" & SLD012 > 24 ~ NA_real_, # General plausibility check
SurveyCycle == "2015-2016" ~ SLD012,

# For the 2017-2018 cycle, calculate the weighted average
SurveyCycle == "2017-2018" ~ {
  # Handle the special codes first
  weekday_sleep <- ifelse(SLD012 > 24, NA, SLD012)
  weekend_sleep <- ifelse(SLD013 > 24, NA, SLD013)

  # Calculate the weighted average
  ((weekday_sleep * 5) + (weekend_sleep * 2)) / 7
},

# A fall-back for any other case
TRUE ~ NA_real_
)
) %>%
select(-c("SLD010H", "SLD012", "SLD013", "SurveyCycle"))

# Final adjustments pre-imputation
imputation_vars_1118 <- imputation_vars_full.4 %>%
mutate(Phys = PAD660 + PAD675,
       Height_m = BMXHT / 100) %>%
select(-c(BMXHT, PAD660, PAD675)) %>%
rename(
  ID = SEQN,
  PSU = SDMVPSU,
  Strata = SDMVSTRA,
  SurvWeight = WTMEC2YR,
  Gender = RIAGENDR,
  Age_yrs = RIDAGEYR,
  Race = RIDRETH3,
  FamIncPov_Ratio = INDFMPIR,
  Bfp_perc = DXDTOPF,
  WaistCircum_cm = BMXWAIST,
  Hba1c_perc = LBXGH,
  RightArmLean_g = DXDRAL,
  LeftArmLean_g = DXDLAL,
  RightLegLean_g = DXDRLL,
  LeftLegLean_g = DXDLLE,
  HEI = score
)

```

7.4 Appendix D: Multiple Imputation

```

# ----- #
# Multiple imputation using Amelia
# ----- #

# Imputation

# Good amount of missingness for Amelia to handle
colSums(is.na(imputation_vars_1118))

# Setting realistic bounds for continuous variables for imputation
bounds_matrix_1118 <- matrix(
  c(
    which(names(imputation_vars_1118) == "FamIncPov_Ratio"), 0, 5, # Family income to poverty ratio
    which(names(imputation_vars_1118) == "Bfp_perc"), 5, 80, # Total Body Fat (%)
    which(names(imputation_vars_1118) == "WaistCircum_cm"), 40, 250, # Waist Circumference (cm)
    which(names(imputation_vars_1118) == "Hba1c_perc"), 3.5, 20.0, # HbA1c (%)
  )
)

```

```

which(names(imputation_vars_1118) == "Height_m"), 1.2, 2.2, # Height (m)
which(names(imputation_vars_1118) == "RightArmLean_g"), 500, 15000, # R Arm Lean (g)
which(names(imputation_vars_1118) == "LeftArmLean_g"), 500, 15000, # L Arm Lean (g)
which(names(imputation_vars_1118) == "RightLegLean_g"), 500, 15000, # R Leg Lean (g)
which(names(imputation_vars_1118) == "LeftLegLean_g"), 500, 15000, # L Leg Lean (g)
which(names(imputation_vars_1118) == "Phys"), 0, 2000, # Moderate + Vigorous Recreational Physical
# Activity (mins)
which(names(imputation_vars_1118) == "AvgDailyDrinks"), 0, 20, # Average # of alcoholic drinks per day
# for the last year
which(names(imputation_vars_1118) == "HEI"), 0, 100, # Healthy Eating Index (HEI)
which(names(imputation_vars_1118) == "AvgNightlySleep"), 2, 15 # Average hours of sleep per night
),
ncol = 3,
byrow = TRUE
)

# Run Amelia
amelia_out_1118 <- amelia(imputation_vars_1118, m = 30,
                          idvars = c("ID", "PSU", "Strata", "SurvWeight"),
                          noms = c("Gender", "Race", "LifetimeDrinkerFlag", "Smoking_Status"),
                          bounds = bounds_matrix_1118,
                          parallel = "multicore",
                          ncpus = 15) # Utilises more CPU cores to speed up processing. Set to # of cores - 1.

# Feature engineering to form ALMI, alcohol status, and HbA1c outcome variables
# Also removing redundant columns
imputed_1118 <- lapply(amelia_out_1118$imputations, function(df) {
  df %>%
  # ALMI
  mutate(
    Alm_kg = (RightArmLean_g + LeftArmLean_g + RightLegLean_g + LeftLegLean_g) / 1000,
    Almi = Alm_kg / (Height_m^2),
  ) %>%

  # Creating Alcohol_Status variable

  # Need to first ensure AvgDailyDrinks is 0 when LifetimeDrinkerFlag is "Never" again.
  # Imputation will have introduced instances where this does not occur due to
  # missing data in LifetimeDrinkerFlag

  mutate(AvgDailyDrinks = ifelse(LifetimeDrinkerFlag == "Never", 0, AvgDailyDrinks)) %>%

  mutate(
    Alcohol_Status = factor(case_when(

      # Condition 1: Never Drinkers
      LifetimeDrinkerFlag == "Never" ~ "Never Drinker",

      # Condition 2: Former Drinkers
      # They have drunk in their life ("Ever") but their current average is 0
      LifetimeDrinkerFlag == "Ever" & AvgDailyDrinks == 0 ~ "Former Drinker",

      # Condition 3: Heavy Drinkers
      Gender == "Female" & AvgDailyDrinks > 1 ~ "Heavy Drinker",
      Gender == "Male" & AvgDailyDrinks > 2 ~ "Heavy Drinker",

      # Condition 4: Moderate Drinkers
      # Any remaining person with AvgDailyDrinks > 0 is a moderate drinker
      AvgDailyDrinks > 0 ~ "Moderate Drinker"

    )),
    # Set the order of the factor levels for clear regression output,
    # with "Never Drinker" as the clean reference category.
    levels = c("Never Drinker", "Former Drinker", "Moderate Drinker", "Heavy Drinker"))
  )

```

```

) %>%

mutate(
  # Outcome for Onset models: At Risk or Worse
  # Ensure this is numeric 0/1
  at_risk_or_worse = ifelse(HbA1c_perc >= 5.7, 1, 0),

  # Outcome for Progression models: Has Diabetes
  is_diabetic = ifelse(HbA1c_perc >= 6.5, 1, 0)
) %>%

select(-c(RightArmLean_g, LeftArmLean_g, RightLegLean_g, LeftLegLean_g, Alm_kg, AvgDailyDrinks,
          LifetimeDrinkerFlag))
})

```

7.5 Appendix E: Exploratory Data Analysis

```

# ----- #
# Exploratory data analysis prior to modelling
# ----- #

# Creating a custom theme
theme_dissertation <- function() {
  theme_minimal() + # Start with minimal theme
  theme(
    # --- Plot Border ---
    # Allow for border around the figure but not around the plotting area
    plot.background = element_rect(color = "black", linewidth = 1),
    panel.border = element_blank(),

    # --- Axis Lines ---
    # Add solid black axis lines for x and y
    axis.line = element_line(color = "black", linewidth = 0.5),

    # --- Gridlines ---
    # Dotted horizontal gridlines, no vertical gridlines
    panel.grid.major.x = element_blank(),
    panel.grid.minor.x = element_blank(),
    panel.grid.major.y = element_line(color = "grey80", linetype = "dotted"),
    panel.grid.minor.y = element_blank(),

    # --- Legend ---
    legend.title = element_text(face = "bold", size = 12),
    legend.text = element_text(size = 11),

    # --- Text Elements ---
    # Set the font sizes for titles and axes
    plot.title = element_text(size = 9, face = "bold", hjust = 0),
    plot.subtitle = element_text(size = 7, hjust = 0),
    axis.title = element_text(size = 8),
    axis.text = element_text(size = 7, color = "black")
  )
}

# Storing imputation 1
impl <- imputed_1118$impl

# --- Distributions of the outcome in continuous form (HbA1c) and the main predictor of interest (ALMI) --- #

# Using all imputations for these
# Bind all 30 imputed datasets into one large data frame
all_imputed_data <- bind_rows(imputed_1118, .id = "imputation_id")

```

```

# Density plot for HbA1c
gluc.dens <- ggplot(all_imputed_data, aes(x = Hba1c_perc, group = imputation_id)) +
  geom_density(
    data = all_imputed_data %>% filter(imputation_id != "imp1"),
    alpha = .5,
    colour = "darkblue"
  ) +
  geom_density(
    data = all_imputed_data %>% filter(imputation_id == "imp1"),
    linewidth = 1,
    colour = "darkblue"
  ) +
  labs(title = "Distribution of Glycated Haemoglobin (HbA1c)", x = "HbA1c (%)", y = "Density") +
  theme_dissertation()

# Density plot for ALMI (Predictor)
muscle.dens <- ggplot(all_imputed_data, aes(x = Almi, group = imputation_id)) +
  geom_density(
    data = all_imputed_data %>% filter(imputation_id != "imp1"),
    alpha = .5,
    colour = "darkblue"
  ) +
  geom_density(
    data = all_imputed_data %>% filter(imputation_id == "imp1"),
    linewidth = 1,
    colour = "darkblue"
  ) +
  labs(title = "Distribution of Appendicular Lean Mass Index (ALMI)", x = "ALMI (kg/m²)", y = "Density") +
  theme_dissertation()

# --- Frequency of categorical variables --- #

# Bar chart for Gender predictor
ggplot(imp1, aes(x = as.factor(Gender))) +
  geom_bar(fill = "steelblue", color = "black", linewidth = 1) +
  labs(title = "Frequency of Genders", x = "Gender", y = "Count") +
  theme_dissertation()

# Bar chart for the onset model outcome 'at_risk_or_worse'
ggplot(imp1, aes(x = as.factor(at_risk_or_worse))) +
  geom_bar(fill = "steelblue", color = "black", linewidth = 1) +
  scale_x_discrete(labels=c("0" = "Healthy", "1" = "Dysglycemia")) +
  labs(title = "Frequency of Dysglycemia Status", x = "Blood Glucose Status", y = "Count") +
  theme_dissertation()

# Bar chart for the progression model outcome 'is_diabetic'
ggplot(imp1, aes(x = as.factor(is_diabetic))) +
  geom_bar(fill = "steelblue", color = "black", linewidth = 1) +
  scale_x_discrete(labels=c("0" = "Non-Diabetic", "1" = "Diabetic")) +
  labs(title = "Frequency of Diabetic Status", x = "Blood Glucose Status", y = "Count") +
  theme_dissertation()

# Bar chart for Alcohol_Status predictor
ggplot(imp1, aes(x = as.factor(Alcohol_Status))) +
  geom_bar(fill = "steelblue", color = "black", linewidth = 1) +
  labs(title = "Frequency of Alcohol Status Categories", x = "Alcohol Status", y = "Count") +
  theme_dissertation()

# Bar chart for Smoking_Status predictor
ggplot(imp1, aes(x = as.factor(Smoking_Status))) +
  geom_bar(fill = "steelblue", color = "black", linewidth = 1) +
  labs(title = "Frequency of Smoking Status Categories", x = "Smoking Status", y = "Count") +
  theme_dissertation()

```

```
# --- Explore how relationships differ across categories using box plots --- #

# Box plot of HbA1c by Sex
ggplot(imp1, aes(x = Gender, y = Hba1c_perc, fill = Gender)) +
  geom_boxplot() +
  scale_x_discrete(labels=c("1" = "Male", "2" = "Female")) +
  labs(title = "HbA1c by Sex", x = "Sex", y = "HbA1c (%)") +
  theme_dissertation()

# Box plot of ALMI by Sex
ggplot(imp1, aes(x = Gender, y = Almi, fill = Gender)) +
  geom_boxplot() +
  scale_x_discrete(labels=c("1" = "Male", "2" = "Female")) +
  labs(title = "ALMI by Sex", x = "Sex", y = "ALMI (kg/m²)") +
  theme_dissertation()

# Box plot of HbA1c by Race/Ethnicity
ggplot(imp1, aes(x = Race, y = Hba1c_perc, fill = Race)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) + # Rotate x-axis labels for readability
  labs(title = "HbA1c by Race/Ethnicity", x = "Race/Ethnicity", y = "HbA1c (%)") +
  theme_dissertation()

# Box plot of ALMI by Race/Ethnicity
ggplot(imp1, aes(x = Race, y = Almi, fill = Race)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "ALMI by Race/Ethnicity", x = "Race/Ethnicity", y = "ALMI (kg/m²)") +
  theme_dissertation()

# --- Visually inspect the relationship between ALMI and HbA1c --- #

# Scatter plot of HbA1c vs. ALMI
ggplot(imp1, aes(x = Almi, y = Hba1c_perc)) +
  geom_point(alpha = 0.4) + # Use alpha for transparency to see dense areas
  geom_smooth(method = "lm", color = "red") + # Add a linear model trend line
  labs(title = "HbA1c vs. ALMI",
       x = "Appendicular Lean Mass Index (kg/m²)",
       y = "Glycated Haemoglobin (HbA1c %)") +
  theme_dissertation()

# Scatter plot of HbA1c vs. ALMI by Gender
gluc.musc.scats <- ggplot(imp1, aes(x = Almi, y = Hba1c_perc, colour=Gender)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", linewidth = 1) +
  scale_colour_manual(values = c("Male" = "#00BFC4", "Female" = "#F8766D")) +
  labs(title = "Glycated Haemoglobin vs. Appendicular Lean Mass Index by Gender",
       x = "ALMI (kg/m²)",
       y = "HbA1c (%)") +
  theme_dissertation()
```

7.6 Appendix F: Modelling

```
# ----- #
# Model Fitting with Stepwise Addition of Covariates
# ----- #

# Survey design list
design_list <- lapply(imputed_1118, function(dataset) {
  svydesign(
    ids = ~PSU,
    strata = ~Strata,
```

```

    weights = ~SurvWeight,
    nest = TRUE,
    data = dataset
  )
})

# Normal vs Elevated - Onset

# at_risk_or_worse ~ Almi
logistic_onset <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset <- MIcombine(logistic_onset)
summary(pooled_logistic_onset)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race
logistic_onset.2 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.2 <- MIcombine(logistic_onset.2)
summary(pooled_logistic_onset.2)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc
logistic_onset.3 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.3 <- MIcombine(logistic_onset.3)
summary(pooled_logistic_onset.3)

# First model with this FamIncPov_Ratio

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm
logistic_onset.4 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.4 <- MIcombine(logistic_onset.4)
summary(pooled_logistic_onset.4)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio
logistic_onset.5 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.5 <- MIcombine(logistic_onset.5)
summary(pooled_logistic_onset.5)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys
logistic_onset.6 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.6 <- MIcombine(logistic_onset.6)
summary(pooled_logistic_onset.6)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
logistic_onset.7 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.7 <- MIcombine(logistic_onset.7)

```

```

summary(pooled_logistic_onset.7)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI +
# Alcohol_Status
logistic_onset.8 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
    + Alcohol_Status,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.8 <- MIcombine(logistic_onset.8)
summary(pooled_logistic_onset.8)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI +
# Alcohol_Status + Smoking_Status
logistic_onset.9 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
    + Alcohol_Status + Smoking_Status,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.9 <- MIcombine(logistic_onset.9)
summary(pooled_logistic_onset.9)

# at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI +
# Alcohol_Status + Smoking_Status + AvgNightlySleep
logistic_onset.10 <- lapply(design_list, function(design_obj) {
  svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
    + Alcohol_Status + Smoking_Status + AvgNightlySleep,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_onset.10 <- MIcombine(logistic_onset.10)
summary(pooled_logistic_onset.10)

# ALMI not significant when moving from normal to elevated HbA1c

# Non-Diabetic vs Diabetic - Progression

# is_diabetic ~ Almi
logistic_progression <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression <- MIcombine(logistic_progression)
summary(pooled_logistic_progression)

# is_diabetic ~ Almi + Age_yrs + Gender + Race
logistic_progression.2 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.2 <- MIcombine(logistic_progression.2)
summary(pooled_logistic_progression.2)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc
logistic_progression.3 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.3 <- MIcombine(logistic_progression.3)
summary(pooled_logistic_progression.3)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm
logistic_progression.4 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm,
    design = design_obj, family = quasibinomial)
})

```

```

})
pooled_logistic_progression.4 <- MIcombine(logistic_progression.4)
summary(pooled_logistic_progression.4)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio
logistic_progression.5 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.5 <- MIcombine(logistic_progression.5)
summary(pooled_logistic_progression.5)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys
logistic_progression.6 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.6 <- MIcombine(logistic_progression.6)
summary(pooled_logistic_progression.6)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
logistic_progression.7 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.7 <- MIcombine(logistic_progression.7)
summary(pooled_logistic_progression.7)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI +
# Alcohol_Status
logistic_progression.8 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
    + Alcohol_Status,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.8 <- MIcombine(logistic_progression.8)
summary(pooled_logistic_progression.8)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI +
# Alcohol_Status + Smoking_Status
logistic_progression.9 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
    + Alcohol_Status + Smoking_Status,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.9 <- MIcombine(logistic_progression.9)
summary(pooled_logistic_progression.9)

# is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI +
# Alcohol_Status + Smoking_Status + AvgNightlySleep
logistic_progression.10 <- lapply(design_list, function(design_obj) {
  svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm + FamIncPov_Ratio + Phys + HEI
    + Alcohol_Status + Smoking_Status + AvgNightlySleep,
    design = design_obj, family = quasibinomial)
})
pooled_logistic_progression.10 <- MIcombine(logistic_progression.10)
summary(pooled_logistic_progression.10)

# ALMI IS significant when moving from non-diabetic to diabetic HbA1c

```

7.7 Appendix G: Diagnostics

```
# ----- #
# Model Diagnostics
# ----- #

# --- Archer-Lemeshow Goodness-of-Fit Test --- #

# Goodness-of-fit function
svy_archer_lemeshow <- function(model, design, groups = 10) {
  residuals <- residuals(model, type = "response")
  fitted_values <- fitted(model)

  fitted_groups <- cut(fitted_values,
                      breaks = quantile(fitted_values, probs = seq(0, 1, 1 / groups), na.rm = TRUE),
                      include.lowest = TRUE)

  design <- update(design,
                  resids = residuals,
                  fitted_groups = fitted_groups)

  gof_model <- svyglm(resids ~ fitted_groups, design = design)

  regTermTest(gof_model, ~fitted_groups)
}

# Obtaining a p-value diagnosing the fit of each pooled model
# p > 0.05 indicates good fit

# Onset

# --- Step 1: Organize all un-pooled model lists into a single master list ---

all_onset_model_lists <- list(
  logistic_onset,
  logistic_onset.2,
  logistic_onset.3,
  logistic_onset.4,
  logistic_onset.5,
  logistic_onset.6,
  logistic_onset.7,
  logistic_onset.8,
  logistic_onset.9,
  logistic_onset.10
)

# Create vector of model names
model_names <- paste("Model", 1:10)

# --- Step 2: Initialize an array to store the final p-values ---
onset.gof.pvalue <- c()

# --- Step 3: Loop through each model list, run the test, and store the result ---

for (i in 1:10) {

  cat("Processing", model_names[i], "...\\n")

  # Select the current list of model fits
  onset.model.list <- all_onset_model_lists[[i]]

  # Initialize an empty vector for F-statistics for the current model
  onset.f.statistics <- c()

  # Loop through each imputation for the current model
```

```

for (j in 1:length(onset.model.list)) {
  current_model <- onset.model.list[[j]]
  current_design <- design_list[[j]]

  # Run the goodness-of-fit test
  gof_test_result <- svy_archer_lemeshow(current_model, current_design)

  # Extract and store the F-statistic
  onset.f.statistics[j] <- gof_test_result$Ftest
}

# Numerator degrees of freedom (df1)
df1 <- gof_test_result$df

# Pool the F-statistics for the current model
pooled_gof_test <- micombine.F(onset.f.statistics, df1 = df1)

# Extract the final pooled p-value
final_p_value <- pooled_gof_test[2]

# Add the result to array
onset.gof.pvalue[i] = final_p_value
}

# Progression

# --- Step 1: Organize all unpooled model lists into a single master list ---

all_prog_model_lists <- list(
  logistic_progression,
  logistic_progression.2,
  logistic_progression.3,
  logistic_progression.4,
  logistic_progression.5,
  logistic_progression.6,
  logistic_progression.7,
  logistic_progression.8,
  logistic_progression.9,
  logistic_progression.10
)

# --- Step 2: Initialize an array to store the final p-values ---
prog.gof.pvalue <- c()

# --- Step 3: Loop through each model list, run the test, and store the result ---

for (i in 1:10) {

  cat("Processing", model_names[i], "...\\n")

  # Select the current list of model fits
  prog.model.list <- all_prog_model_lists[[i]]

  # Initialize an empty vector for F-statistics for the current model
  prog.f.statistics <- c()

  # Loop through each imputation for the current model
  for (j in 1:length(prog.model.list)) {
    current_model <- prog.model.list[[j]]
    current_design <- design_list[[j]]

    # Run the goodness-of-fit test
    gof_test_result <- svy_archer_lemeshow(current_model, current_design)

```

```

    # Extract and store the F-statistic
    prog.f.statistics[j] <- gof_test_result$Ftest
  }

  # Numerator degrees of freedom (df1)
  df1 <- gof_test_result$df

  # Pool the F-statistics for the current model
  pooled_gof_test <- micombine.F(prog.f.statistics, df1 = df1)

  # Extract the final pooled p-value
  final_p_value <- pooled_gof_test[2]

  # Add the result to array
  prog.gof.pvalue[i] = final_p_value
}

# --- Creating lists of mean McFadden Pseudo R2 values for each pooled model --- #

# Onset

# Function for finding the mean McFadden Pseudo R2 from list of models
mean_fit_stat <- function(model_list) {
  # Apply fit.svyglm to each model
  fit_results <- lapply(model_list, fit.svyglm)

  # Extract the first element from each result
  fit_values <- sapply(fit_results, function(x) x[1])

  # Compute and return the mean
  mean(fit_values)
}

# Create the names of the 9 lists as strings
onset.model.names <- c("logistic_onset", paste0("logistic_onset.", 2:10))

# Apply mean_fit_stat to each model list by name
onset.pseudo.r2.list <- sapply(onset.model.names, function(name) {
  model_list <- get(name)
  mean_fit_stat(model_list)
})

# Show results
print(onset.pseudo.r2.list)

# Progression

# Create the names of the 9 lists as strings
prog.model.names <- c("logistic_progression", paste0("logistic_progression.", 2:10))

# Apply mean_fit_stat to each model list by name
prog.pseudo.r2.list <- sapply(prog.model.names, function(name) {
  model_list <- get(name)
  mean_fit_stat(model_list)
})

# Show results
print(prog.pseudo.r2.list)

# Removing array element names for each pseudo R2 list
names(onset.pseudo.r2.list) <- NULL
names(prog.pseudo.r2.list) <- NULL

```

```

# The computed diagnostic statistics are compiled into concise summary tables in script 7 (summaries)

# --- Residual Diagnostics --- #

# The response variable is binary.
# The residuals are not expected to be normally distributed.
# Survey-weighted estimation can distort residual behaviour.

# binnedplot in the arm package is ideal for this case

# Seek to check binned residuals of the fully adjusted onset model and parsimonious progression model
# for all imputations to assess model fit and if there are any significant
# differences between imputed datasets

# Will only include imputation 1 in thesis to save space

# Onset

# Loop through the imputations for the onset model
for (i in 1:30) {

  model_fit <- logistic_onset.10[[i]]

  binnedplot(
    fitted(model_fit),
    residuals(model_fit, type = "response"),
    nclass = NULL,
    xlab = "Expected Values",
    ylab = "Average residual",
    main = paste("Onset Model - Imputation", i), # Add a dynamic title
    cex.pts = 0.8,
    col.pts = 1,
    col.int = "gray"
  )
}

# Progression

# Loop through the imputations for the progression model
for (i in 1:30) {

  model_fit <- logistic_progression.4[[i]]

  binnedplot(
    fitted(model_fit),
    residuals(model_fit, type = "response"),
    nclass = NULL,
    xlab = "Expected Values",
    ylab = "Average residual",
    main = paste("Progression Model - Imputation", i),
    cex.pts = 0.8,
    col.pts = 1,
    col.int = "gray"
  )
}

# Residuals fine for onset model - some patterns evident in progression model
# No significant differences between imputations

# Now creating binned component + residual plots to justify the choice of linear
# covariates in both models

# Only using imputation 1 here

```

```

# Onset

# Get the working residuals and the specific predictor
onset.model.fit <- logistic_onset.10$impl
onset.y.outcome <- onset.model.fit$survey.design$variables$at_risk_or_worse
onset.mu.fitted <- fitted(onset.model.fit)
onset.working.residuals <- (onset.y.outcome - onset.mu.fitted) / onset.mu.fitted

onset.almi.data <- onset.model.fit$survey.design$variables$Almi
onset.age.data <- onset.model.fit$survey.design$variables$Age_yrs
onset.bfp.data <- onset.model.fit$survey.design$variables$Bfp_perc
onset.waistcircum.data <- onset.model.fit$survey.design$variables$WaistCircum_cm
onset.pir.data <- onset.model.fit$survey.design$variables$FamIncPov_Ratio
onset.phys.data <- onset.model.fit$survey.design$variables$Phys
onset.hei.data <- onset.model.fit$survey.design$variables$HEI
onset.sleep.data <- onset.model.fit$survey.design$variables$AvgNightlySleep

# Create the binned plot of residuals vs. the predictor
par(mfrow = c(2, 4))
binnedplot(x = onset.almi.data, y = onset.working.residuals,
           xlab = "ALMI (kg/m²)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. ALMI")

binnedplot(x = onset.age.data, y = onset.working.residuals,
           xlab = "Age (years)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Age")

binnedplot(x = onset.bfp.data, y = onset.working.residuals,
           xlab = "Body Fat Percentage (%)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Body Fat Percentage")

binnedplot(x = onset.waistcircum.data, y = onset.working.residuals,
           xlab = "Waist Circumference (cm)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Waist Circumference")

binnedplot(x = onset.pir.data, y = onset.working.residuals,
           xlab = "Poverty Income Ratio",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Poverty Income Ratio")

binnedplot(x = onset.phys.data, y = onset.working.residuals,
           xlab = "Physical Activity (mins/day)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Physical Activity")

binnedplot(x = onset.hei.data, y = onset.working.residuals,
           xlab = "Healthy Eating Index",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Healthy Eating Index")

binnedplot(x = onset.sleep.data, y = onset.working.residuals,
           xlab = "Average Nightly Sleep (hrs)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Average Nightly Sleep")

par(mfrow = c(2, 2))

# Prog
prog.model.fit <- logistic_progression.4$impl
prog.y.outcome <- prog.model.fit$survey.design$variables$is_diabetic

```

```

prog.mu.fitted <- fitted(prog.model.fit)
prog.working.residuals <- (prog.y.outcome - prog.mu.fitted) / prog.mu.fitted

prog.almi.data <- prog.model.fit$survey.design$variables$Almi
prog.age.data <- prog.model.fit$survey.design$variables$Age_yrs
prog.bfp.data <- prog.model.fit$survey.design$variables$Bfp_perc
prog.waistcircum.data <- prog.model.fit$survey.design$variables$WaistCircum_cm

# Plotting

binnedplot(x = prog.almi.data, y = prog.working.residuals,
           xlab = "ALMI (kg/m²)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. ALMI")

binnedplot(x = prog.age.data, y = prog.working.residuals,
           xlab = "Age (years)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Age")

binnedplot(x = prog.bfp.data, y = prog.working.residuals,
           xlab = "Body Fat Percentage (%)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Body Fat Percentage")

binnedplot(x = prog.waistcircum.data, y = prog.working.residuals,
           xlab = "Waist Circumference (cm)",
           ylab = "Average Working Residual",
           main = "Binned Residuals vs. Waist Circumference")

par(mfrow = c(1, 1))

# No evidence that higher order terms are needed

# ---- Checking for influential points ---- #

# Using survey weighted cook's distance plots

# Onset

# Onset model 4 shows most cooks distances cluster below 10
# It is useful to investigate the outlier observations and how the model performs without them
# We do not want conclusions to depend on a small amount of influential points

# Finding and plotting cooks distances for onset model 10 on imputation 1
# Crucially, the results are very similar on other imputations, so only considering imputation 1 here is okay
# Earlier residual checks also showed no significant differences between imputations
onset.cooks <- svyCooksD(logistic_onset.10$imp1, doplot=TRUE)
# Saving indices of those points with the top 1% cooks distance
onset_cooks_threshold <- quantile(onset.cooks, probs = 0.99)
onset_influential_indices <- which(onset.cooks > onset_cooks_threshold)

# Creating a data frame with these influential observations
# Nothing immediately stands out with these
onset_influential_obs <- design_list$imp1$variables[onset_influential_indices,]
onset_influential_obs$CooksD <- onset.cooks[onset_influential_indices]

# Seek to refit the model without these points

# Create a new data frame by EXCLUDING the influential rows
onset_reduced_data <- design_list$imp1$variables[-onset_influential_indices, ]

# Create a new survey design object with this reduced dataset
onset_reduced_design <- svydesign(id = ~PSU,

```

```

        strata = ~Strata,
        weights = ~SurvWeight,
        nest = TRUE,
        data = onset_reduced_data)

# Refit the model using the reduced design object
onset_without_influential <- svyglm(at_risk_or_worse ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm,
                                   design = onset_reduced_design,
                                   family = quasibinomial)

# Almi coefficient is still not significant
# This means that removing influential points does not change the findings
summary(onset_without_influential)

# Now checking this for progression model 4

prog.cooks <- svyCooksD(logistic_progression.4$imp1, doplot=TRUE)
# Saving indices of those points with the top 1% cooks distance
prog_cooks_threshold <- quantile(prog.cooks, probs = 0.99)
prog_influential_indices <- which(prog.cooks > prog_cooks_threshold)

# Creating a data frame with these influential observations
# Nothing immediately stands out
prog_influential_obs <- design_list$imp1$variables[prog_influential_indices,]
prog_influential_obs$CooksD <- prog.cooks[prog_influential_indices]

# Refit the model without these points

# Create a new data frame without the influential rows
prog_reduced_data <- design_list$imp1$variables[-prog_influential_indices, ]

# Create a new survey design object with this reduced dataset
prog_reduced_design <- svydesign(id = ~PSU,
                                strata = ~Strata,
                                weights = ~SurvWeight,
                                nest = TRUE,
                                data = prog_reduced_data)

# Refit the model using the reduced design object
prog_without_influential <- svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm,
                                   design = prog_reduced_design,
                                   family = quasibinomial)

# Once again, nothing has significantly changed.
# Can conclude that findings are robust and not dependant on influential outliers.
summary(prog_without_influential)

```

7.7.1 Binned Residual Plots

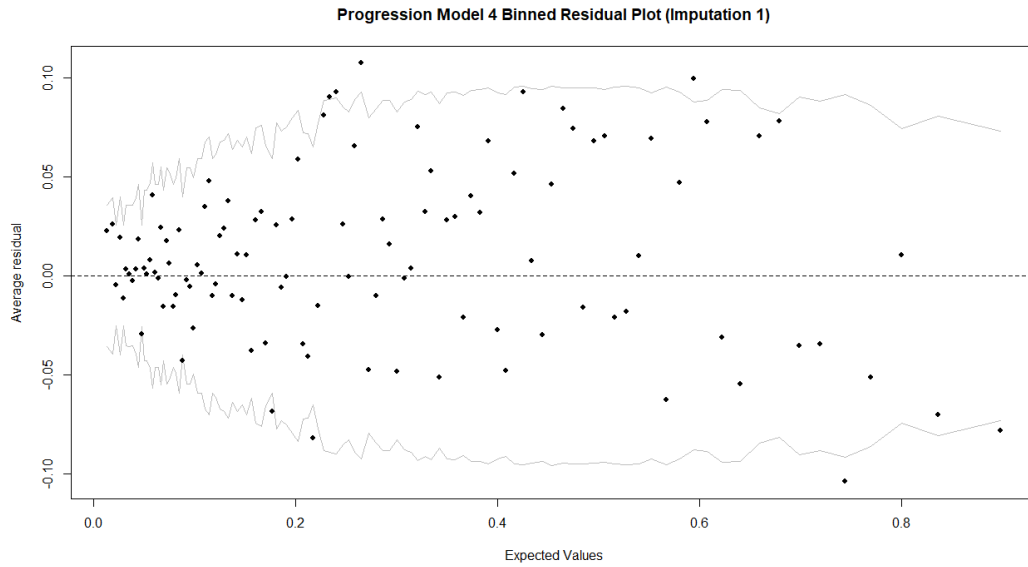


Figure 6: Binned residual plot for Onset Model 10, using the first imputed dataset as a representative example.

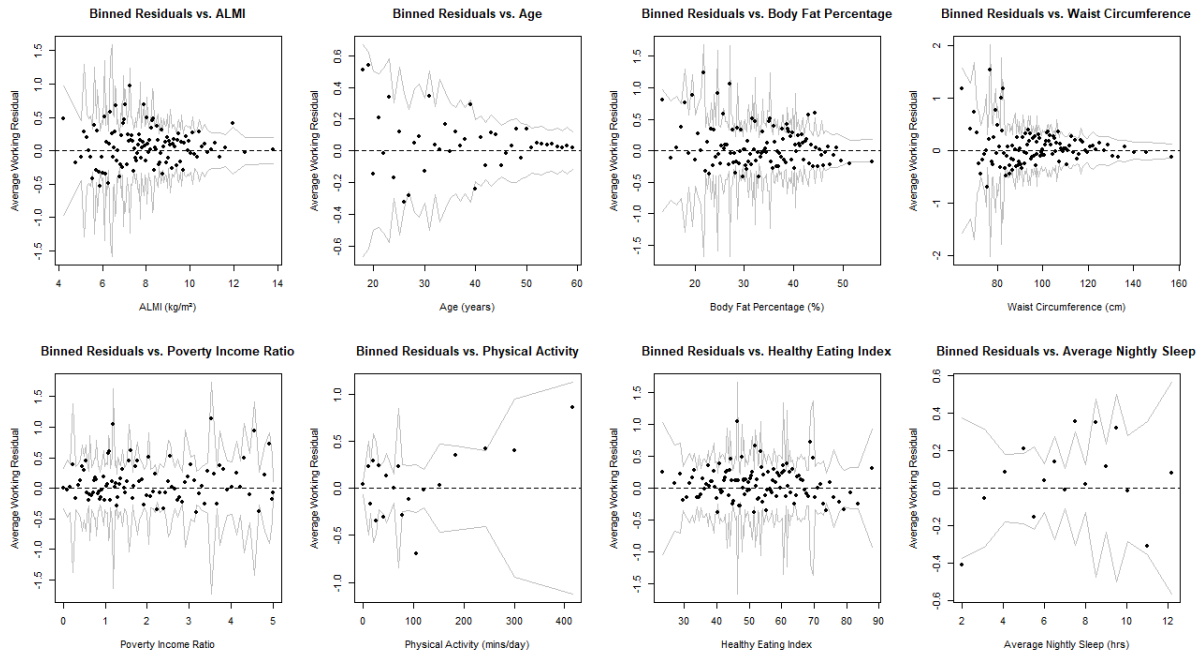


Figure 7: Component-wise binned residual plots for Onset Model 10, used to assess the linearity assumption for each continuous predictor. Plots were generated using the first imputed dataset as a representative example.

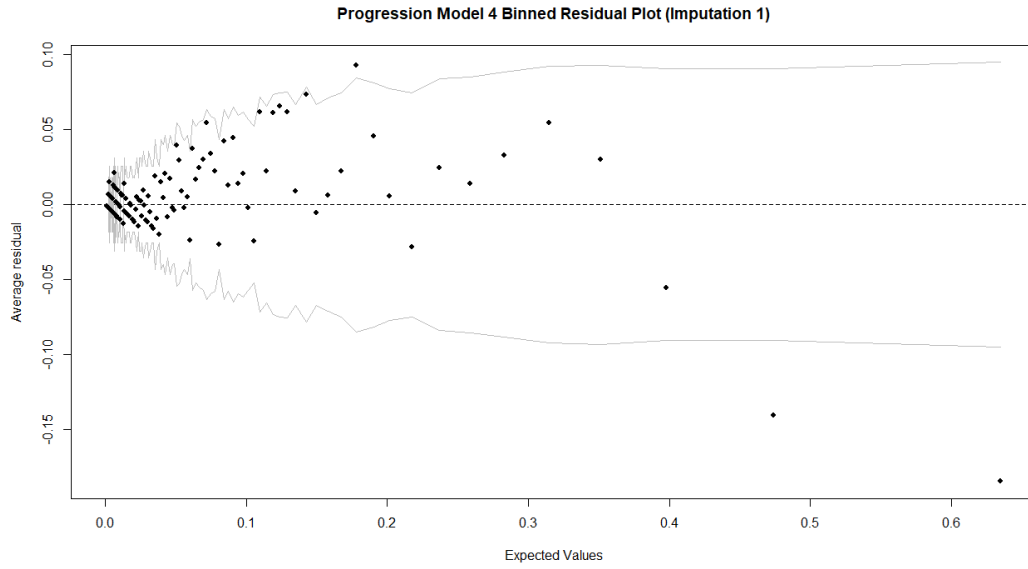


Figure 8: Binned residual plot for Progression Model 4, using the first imputed dataset as a representative example.

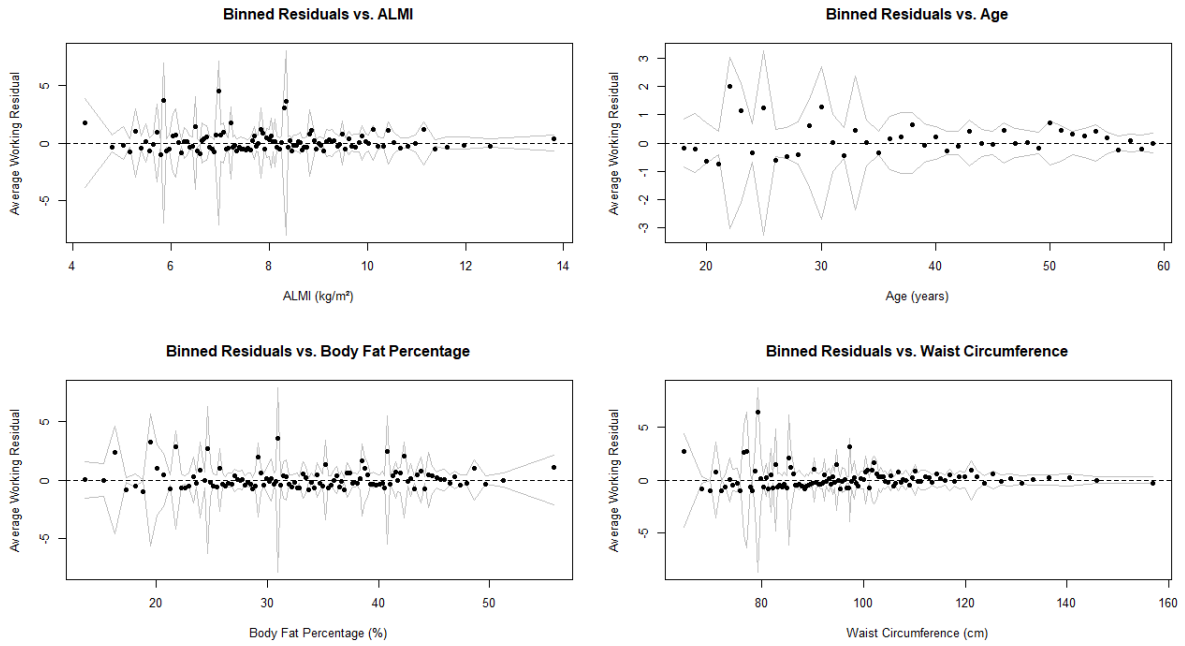


Figure 9: Component-wise binned residual plots for Progression Model 4, used to assess the linearity assumption for each continuous predictor. Plots were generated using the first imputed dataset as a representative example.

7.8 Appendix H: Model Results and Summaries

```
# -----#
# Data & Model summaries
# -----#

# Descriptive statistics table of the final analytical sample

# Seek to obtain the mean and standard deviation of each continuous variable, and
# the count and proportion of each categorical variable from each imputed dataset.
# The survey structure must be taken into account for these tasks using functions
# such as svymean for survey-weighted means and svyvar for survey-weighted variances.

# These statistics will then be pooled and used to produce a concise summary statistics
# table.

# --- Step 1: Define a function to calculate summary stats for ONE dataset ---
get_summary_stats <- function(design) {
  # --- Continuous Variables ---
  continuous_vars <- c("Age_yrs", "Almi", "WaistCircum_cm", "Bfp_perc",
    "FamIncPov_Ratio", "HEI", "Phys", "AvgNightlySleep", "Hba1c_perc")

  cont_summary <- map_df(continuous_vars, ~{
    mean_val <- svymean(as.formula(paste0("~", .x)), design, na.rm = TRUE)
    var_val <- svyvar(as.formula(paste0("~", .x)), design, na.rm = TRUE)
    tibble(
      Variable = .x,
      Mean = as.numeric(coef(mean_val)),
      SD = sqrt(as.numeric(var_val)),
      Type = "Continuous"
    )
  })

  # --- Categorical Variables ---
  categorical_vars <- c("Gender", "Race", "Smoking_Status", "Alcohol_Status", "at_risk_or_worse", "is_diabetic")

  cat_summary <- map_df(categorical_vars, ~{
    tbl <- svytable(as.formula(paste0("~", .x)), design)
    props <- prop.table(tbl)
    tibble(
      Variable = paste0(.x, ", ", ", ", names(tbl)),
      Prop = as.numeric(props),
      Type = "Categorical"
    )
  })

  bind_rows(cont_summary, cat_summary)
}

# --- Step 2: Apply the function to all 30 design objects and pool ---
# map_df runs the function on each design object and stacks the results.
pooled_results <- map_df(design_list, get_summary_stats, .id = "imputation")

# Now, calculate the final pooled estimates by averaging across imputations
final_summary <- pooled_results %>%
  group_by(Variable, Type) %>%
  summarise(
    Pooled_Mean = mean(Mean, na.rm = TRUE),
    Pooled_SD = mean(SD, na.rm = TRUE),
    Pooled_Prop = mean(Prop, na.rm = TRUE),
    .groups = "drop"
  )

# --- Step 3: Format the final table ---
# Get total N from the first imputation (it's the same for all)
```

```

total_n <- nrow(imputed_1118[[1]])

# Format the continuous and categorical results separately
formatted_continuous <- final_summary %>%
  filter(Type == "Continuous") %>%
  mutate(Value = sprintf("%.1f (%.1f)", Pooled_Mean, Pooled_SD)) %>%
  select(Variable, Value)

formatted_categorical <- final_summary %>%
  filter(Type == "Categorical") %>%
  mutate(
    Pooled_N = Pooled_Prop * total_n,
    Value = sprintf("%.Of (%.1f%%)", Pooled_N, Pooled_Prop * 100)
  ) %>%
  select(Variable, Value)

variable_order <- c(
  # Continuous Variables
  "Age (years)", "ALMI (kg/m²)", "HbA1c (%)", "Body Fat Percentage (%)",
  "Waist Circumference (cm)", "Poverty Income Ratio", "Physical Activity (mins/day)",
  "Healthy Eating Index", "Average Nightly Sleep (hours)",
  # Categorical Variables
  "Normal HbA1c", "Elevated HbA1c", "Non-Diabetic", "Diabetic",
  "Male", "Female",
  "Mexican American", "Hispanic", "Non-Hispanic White",
  "Non-Hispanic Black", "Non-Hispanic Asian", "Other Race",
  "Never Drinker", "Former Drinker", "Moderate Drinker", "Heavy Drinker",
  "Never Smoker", "Former Smoker", "Current Smoker"
)

# Combine into the final data frame for the table
final_table_df <- bind_rows(formatted_continuous, formatted_categorical) %>%
  mutate(Variable_Clean = case_when(
    Variable == "Almi" ~ "ALMI (kg/m²)",
    Variable == "HbA1c_perc" ~ "HbA1c (%)",
    Variable == "at_risk_or_worse, 0" ~ "Normal HbA1c",
    Variable == "at_risk_or_worse, 1" ~ "Elevated HbA1c",
    Variable == "is_diabetic, 0" ~ "Non-Diabetic",
    Variable == "is_diabetic, 1" ~ "Diabetic",
    Variable == "Age_yrs" ~ "Age (years)",
    Variable == "Gender, Male" ~ "Male",
    Variable == "Gender, Female" ~ "Female",
    Variable == "Race, Mexican American" ~ "Mexican American",
    Variable == "Race, Other Hispanic" ~ "Hispanic",
    Variable == "Race, Non-Hispanic White" ~ "Non-Hispanic White",
    Variable == "Race, Non-Hispanic Black" ~ "Non-Hispanic Black",
    Variable == "Race, Non-Hispanic Asian" ~ "Non-Hispanic Asian",
    Variable == "Race, Other Race - Including Multi-Racial" ~ "Other Race",
    Variable == "Bfp_perc" ~ "Body Fat Percentage (%)",
    Variable == "WaistCircum_cm" ~ "Waist Circumference (cm)",
    Variable == "FamIncPov_Ratio" ~ "Poverty Income Ratio",
    Variable == "Phys" ~ "Physical Activity (mins/day)",
    Variable == "HEI" ~ "Healthy Eating Index",
    Variable == "Alcohol_Status, Never Drinker" ~ "Never Drinker",
    Variable == "Alcohol_Status, Former Drinker" ~ "Former Drinker",
    Variable == "Alcohol_Status, Moderate Drinker" ~ "Moderate Drinker",
    Variable == "Alcohol_Status, Heavy Drinker" ~ "Heavy Drinker",
    Variable == "Smoking_Status, Never Smoker" ~ "Never Smoker",
    Variable == "Smoking_Status, Former Smoker" ~ "Former Smoker",
    Variable == "Smoking_Status, Current Smoker" ~ "Current Smoker",
    Variable == "AvgNightlySleep" ~ "Average Nightly Sleep (hours)",
    TRUE ~ Variable # Keep original name if no match
  )) %>%
  # Convert the clean Variable name to a factor to enforce order

```

```

mutate(Variable_Clean = factor(Variable_Clean, levels = variable_order)) %>%
# Arrange the data according to the factor levels
arrange(Variable_Clean) %>%
# Select the final columns for the table
select(Variable = Variable_Clean, Value) %>%
# Renaming variable column
rename(Characteristic = Variable)

# --- Create the Table ---
descriptive.tbl.kbl <- kbl(
  final_table_df,
  format = "latex",
  booktabs = TRUE
) %>%
kable_styling(
  font_size = 7,
  full_width = FALSE
) %>%
pack_rows("Glycaemic Status (Onset)", 10, 11) %>%
pack_rows("Diabetes Status (Progression)", 12, 13) %>%
pack_rows("Gender", 14, 15) %>%
pack_rows("Race", 16, 21) %>%
pack_rows("Alcohol Consumption", 22, 25) %>%
pack_rows("Smoking Status", 26, 28) %>%
footnote(
  general = "ALMI: Appendicular Lean Mass Index; HbA1c: Haemoglobin A1c.",
  symbol = "Continuous variables presented as mean (SD), categorical variables presented as n (%).",
  threeparttable = TRUE,
  footnote_as_chunk = TRUE
)

# --- Hierarchical model building tables --- #

# Extracting pooled model ALMI coefficients and confidence intervals
# Also finding the % change of each ALMI coefficient progressing through the models

# Onset

# Model names
pooled.onset.names <- c("pooled_logistic_onset", paste0("pooled_logistic_onset.", 2:10))

# ALMI coefficients
onset.almi.coeffs <- sapply(pooled.onset.names, function(name) {
  model_list <- get(name)
  coef(model_list)["Almi"]
})

# ALMI confidence intervals
onset.almi.confint <- sapply(pooled.onset.names, function(name) {
  model_list <- get(name)
  c(confint(model_list)[2,1], confint(model_list)[2,2])
})

# % changes
onset.percent.changes <- c("N/A") # First % is NA as there is no comparison
for (i in 1:9) {
  onset.percent.changes[i+1] <- ((onset.almi.coeffs[i + 1] - onset.almi.coeffs[i]) / abs(onset.almi.coeffs[i])) * 100
}

# Progression

# Model names
pooled.prog.names <- c("pooled_logistic_progression", paste0("pooled_logistic_progression.", 2:10))

```

```

# ALMI coefficients
prog.almi.coefs <- sapply(pooled.prog.names, function(name) {
  model_list <- get(name)
  coef(model_list)["Almi"]
})

# ALMI confidence intervals
prog.almi.confint <- sapply(pooled.prog.names, function(name) {
  model_list <- get(name)
  c(confint(model_list)[2,1], confint(model_list)[2,2])
})

# % changes
prog.percent.changes <- c("N/A")
for (i in 1:9) {
  prog.percent.changes[i+1] <- ((prog.almi.coefs[i + 1] - prog.almi.coefs[i]) / abs(prog.almi.coefs[i])) * 100
}

# ----- ONSET SUMMARY TABLE ----- #

# Removing names from coefs and confint objects
names(onset.almi.coefs) <- NULL
colnames(onset.almi.confint) <- NULL

# Compute exponentiated confidence intervals
onset.lower.bounds <- exp(onset.almi.confint[1,])
onset.upper.bounds <- exp(onset.almi.confint[2,])

# Combine into a single string column
onset.ci.combined <- paste0("(", sprintf("%.3f", onset.lower.bounds), ", ", sprintf("%.3f", onset.upper.bounds), ")")

# Formatting percentage change column
onset.percent.changes.formatted <- c("N/A")
value <- c()
for (i in 2:10) {
  value[i] <- as.numeric(onset.percent.changes[i])
  onset.percent.changes.formatted[i] = paste0(sprintf("%.2f", value[i]), "%")
}

# Array of covariate changes for summary tables
covariates <- c("Crude", "+ Demographics", "+ Body Fat Percentage (%)", "+ Waist Circumference (cm)",
  "+ Poverty Income Ratio", "+ Physical Activity (mins)", "+ Healthy Eating Index", "+ Alcohol Status",
  "+ Smoking Status", "+ Average Daily Sleep (hrs)")

# Data frame
onset.summary.df <- data.frame(
  Model = paste("Model", 1:10),
  `Covariate Changes` = covariates,
  `ALMI Odds Ratio (OR)` = exp(onset.almi.coefs),
  `95% Confidence Interval (OR)` = onset.ci.combined,
  `ALMI Coefficient (log OR)` = onset.almi.coefs,
  `Percentage Change in Coefficient` = onset.percent.changes.formatted,
  `McFadden Pseudo R^2` = onset.pseudo.r2.list,
  `Archer-Lemeshow GoF Test P-value` = onset.gof.pvalue,
  check.names = FALSE
)

# Creating table
onset.summary.table.kbl <- kbl(
  onset.summary.df,
  format = "latex",
  booktabs = TRUE,
  linesep = "",

```

```

  digits = c(NA, NA, 3, NA, 3, NA, 3, 3),
  align = c("l", "l", "r", "r", "r", "r", "r", "r")
) %>%
  kable_styling(
    font_size = 8
  ) %>%
  add_header_above(
    c("Odds Ratios, Coefficients, Percentage Changes, and Diagnostics" = 8),
    bold = FALSE
  ) %>%
  column_spec(column = 2, width = "15em") %>%
  column_spec(column = 3, width = "4em") %>%
  column_spec(column = 4, width = "7em") %>%
  column_spec(column = 5, width = "5em") %>%
  column_spec(column = 6, width = "5em") %>%
  column_spec(column = 7, width = "5em") %>%
  column_spec(column = 8, width = "5em")

# ----- PROG SUMMARY TABLE ----- #

names(prog.almi.coefs) <- NULL
colnames(prog.almi.confint) <- NULL

# Compute exponentiated confidence intervals
prog.lower.bounds <- exp(prog.almi.confint[1,])
prog.upper.bounds <- exp(prog.almi.confint[2,])

# Combine into a single string column
prog.ci.combined <- paste0("(", sprintf("%.3f", prog.lower.bounds), ", ", sprintf("%.3f", prog.upper.bounds), ")")

# Formatting percentage change column
prog.percent.changes.formatted <- c("N/A")
value <- c()
for (i in 2:10) {
  value[i] <- as.numeric(prog.percent.changes[i])
  prog.percent.changes.formatted[i] = paste0(sprintf("%.2f", value[i]), "%")
}

# Data frame
prog.summary.df <- data.frame(
  Model = paste("Model", 1:10),
  `Covariate Changes` = covariates,
  `ALMI Odds Ratio (OR)` = exp(prog.almi.coefs),
  `95% Confidence Interval (OR)` = prog.ci.combined,
  `ALMI Coefficient (log OR)` = prog.almi.coefs,
  `Percentage Change in Coefficient` = prog.percent.changes.formatted,
  `McFadden Pseudo R^2` = prog.pseudo.r2.list,
  `Archer-Lemeshow GoF Test P-value` = prog.gof.pvalue,
  check.names = FALSE
)

# Creating table
prog.summary.table.kbl <- kbl(
  prog.summary.df,
  format = "latex",
  booktabs = TRUE,
  linesep = "",
  digits = c(NA, NA, 3, NA, 3, NA, 3, 3),
  align = c("l", "l", "r", "r", "r", "r", "r", "r")
) %>%
  kable_styling(
    font_size = 8
  ) %>%

```

```

add_header_above(
  c("Odds Ratios, Coefficients, Percentage Changes, and Diagnostics" = 8),
  bold = FALSE
) %>%
column_spec(column = 2, width = "15em") %>%
column_spec(column = 3, width = "4em") %>%
column_spec(column = 4, width = "7em") %>%
column_spec(column = 5, width = "5em") %>%
column_spec(column = 6, width = "5em") %>%
column_spec(column = 7, width = "5em") %>%
column_spec(column = 8, width = "5em")

# -- VIF values evidencing the masking effect of waist circumference seen when moving from onset
# and progression models 3 to 4 -- #

# Check of survey adjusted VIF values of progression model 4 show high significant multicollinearity
# This is not a problem - it explains the effect that causes the ALMI coefficient
# becoming negative once waist circumference is added to the model.

# Onset

# Model 3
# Design matrix from the model formula and data
X.onset.3 <- model.matrix(logistic_onset.3$imp1$formula, data = design_list$imp1$variables)
# Remove the intercept column
X.onset.3 <- X.onset.3[, -1]
# Extract the survey weights from the design object
weights <- weights(design_list$imp1)
# Run svyvif
onset_vif_values.3 <- svyvif(mobj = logistic_onset.3$imp1, X = X.onset.3, w = weights)
# Print the results
print(onset_vif_values.3$`Intercept adjusted`)

# Model 4
X.onset.4 <- model.matrix(logistic_onset.4$imp1$formula, data = design_list$imp1$variables)
X.onset.4 <- X.onset.4[, -1]
onset_vif_values.4 <- svyvif(mobj = logistic_onset.4$imp1, X = X.onset.4, w = weights)
print(onset_vif_values.4$`Intercept adjusted`)

# Progression

# Model 3
X.prog.3 <- model.matrix(logistic_progression.3$imp1$formula, data = design_list$imp1$variables)
X.prog.3 <- X.prog.3[, -1]
prog_vif_values.3 <- svyvif(mobj = logistic_progression.3$imp1, X = X.prog.3, w = weights)
print(prog_vif_values.3$`Intercept adjusted`)

# Model 4
X.prog.4 <- model.matrix(logistic_progression.4$imp1$formula, data = design_list$imp1$variables)
X.prog.4 <- X.prog.4[, -1]
prog_vif_values.4 <- svyvif(mobj = logistic_progression.4$imp1, X = X.prog.4, w = weights)
print(prog_vif_values.4$`Intercept adjusted`)

# Before waist circumference is added, there is no evidence of severe multicollinearity,
# as seen by the weighted VIF values of progression model 3.
# This further reinforces the findings.

```

7.9 Appendix I: Model Prediction

```

# ----- #
# Model Predictions
# ----- #

```

```

# Using Progression model 4 to find predictions of the probability of becoming diabetic vs ALMI
# Predictions will be made on all 30 imputations and then combined using Rubin's Rules
# Will produce predictions for both male and female because model coefficients suggested
# odds of diabetes were largely different between men and women

# --- Step 1: Define model and data objects ---
m <- 30 # Number of imputations

# Use the first imputed dataset as a template for non-imputed variables and ranges
template_data <- imputed_1118[[1]]

# Calculate the pooled mean for the continuous variables that were involved in imputation
pooled_mean_bfp <- mean(all_imputed_data$Bfp_perc)
pooled_mean_waist <- mean(all_imputed_data$WaistCircum_cm)

# For variables that were NOT imputed, take the value from the template
# This is more efficient as the value is the same across all imputed datasets.
pooled_mean_age <- mean(template_data$Age_yrs)

# Calculate the mode for Race from the template data
pooled_mode_race <- names(which.max(table(template_data$Race)))

# --- Step 2: Create prediction data frames (one for each gender) ---

# Create a sequence of ALMI values to predict over
almi_range <- seq(min(template_data$Almi),
                  max(template_data$Almi),
                  length.out = 1000)

# Base prediction df with average/modal values for all other confounders
predict_df_base <- data.frame(
  Almi = almi_range,
  Age_yrs = pooled_mean_age,
  Race = factor(pooled_mode_race, levels = levels(template_data$Race)),
  Bfp_perc = pooled_mean_bfp,
  WaistCircum_cm = pooled_mean_waist
)

# Create a specific version for Males
predict_df_male <- predict_df_base %>%
  mutate(Gender = factor("Male", levels = levels(template_data$Gender)))

# Create a specific version for Females
predict_df_female <- predict_df_base %>%
  mutate(Gender = factor("Female", levels = levels(template_data$Gender)))

# Combine into a single newdata object for easier looping
predict_df_combined <- rbind(predict_df_male, predict_df_female)

# --- Step 3: Predict on ALL Imputations ---
# Loop through each of the 30 models and get predictions.

prediction_list <- lapply(logistic_progression.4, function(model) {

  # Get predictions and the standard error for each in the logit scale
  # This prevents negative probability confidence intervals
  preds <- predict(model, newdata = predict_df_combined, type = "link", se.fit = TRUE)

  # Combine the predictors and the predictions into a single dataframe
  imputation_results <- cbind(predict_df_combined, as.data.frame(preds))

  return(imputation_results)
})

```

```

# --- Step 4: Pool the Results Using Rubin's Rules ---
# Now combine the 30 sets of predictions into one final result.

# First, stack all the data frames from the list into one big data frame,
# adding an ID for each imputation.
all_predictions <- bind_rows(prediction_list, .id = "imputation_id")

# Now, group by each prediction point and apply Rubin's rules
plot_data_pooled <- all_predictions %>%
  # Rename columns for clarity
  rename(logit_pred = link, se = SE) %>%
  group_by(Almi, Age_yrs, Gender, Race, Bfp_perc, WaistCircum_cm) %>%
  summarise(
    # Pool the logit predictions
    pooled_logit = mean(logit_pred),
    within_imputation_var = mean(se^2),
    between_imputation_var = var(logit_pred),
    total_variance = within_imputation_var + (1 + 1/m) * between_imputation_var,
    pooled_se_logit = sqrt(total_variance),
    .groups = 'drop'
  ) %>%
  # Back-transform to the Probability Scale
  mutate(
    # Back-transform the point estimate
    pooled_prevalence = plogis(pooled_logit),

    # Calculate the CI on the logit scale
    lower_ci_logit = pooled_logit - 1.96 * pooled_se_logit,
    upper_ci_logit = pooled_logit + 1.96 * pooled_se_logit,

    # Back-transform the CI bounds to the probability scale
    lower_ci = plogis(lower_ci_logit),
    upper_ci = plogis(upper_ci_logit)
  )

# --- Step 5: Create the final plot with two lines ---

pred.plot <- ggplot(plot_data_pooled, aes(x = Almi, y = pooled_prevalence, color = Gender, fill = Gender)) +
  # Add the prediction lines
  geom_line(linewidth = 1.2) +

  # Add the confidence ribbons
  geom_ribbon(aes(ymin = lower_ci, ymax = upper_ci), alpha = 0.2, linetype = "blank") +

  # Customize colours and labels
  scale_color_manual(values = c("Male" = "blue", "Female" = "red")) +
  scale_fill_manual(values = c("Male" = "blue", "Female" = "red")) +

  labs(
    title = "Predicted Prevalence of Diabetes by Muscle Mass (ALMI) and Gender",
    subtitle = "Pooled across 30 imputations for a person with average age and body composition",
    x = "Appendicular Lean Mass Index (kg/m²)",
    y = "Predicted Prevalence of Diabetes"
  ) +
  theme_dissertation()

```

7.10 Appendix J: Cross Validation

```

# ----- #
# K-Fold Cross Validation
# ----- #

```

```

# Seek to choose one imputation and conduct k-fold cross validation parsimonious
# Progression model 4

# Goal is to assess how well the model performance would be when trained using other datasets

# Using 5-fold cross validation for 10 repeats

# Using a custom algorithm for flexibility

# The algorithm will calculate the optimal threshold by maximising Youden's J statistic
# This will be done using the training data for each loop

# Also calculating the sensitivity and specificity for each loop for further helpful diagnostics

# Very important to take survey weights into account in this process

# Creating a duplicate of imputation 1
imp1.2 <- imputed_1118$imp1

# Parameters
k <- 5
n_repeats <- 10

# Setting seed for reproducibility
set.seed(123)

# Data frame to store summary results per repeat
repeat_summary <- data.frame(
  `repeat` = integer(n_repeats),
  mean_weighted_auc = numeric(n_repeats),
  mean_optimal_threshold = numeric(n_repeats),
  mean_weighted_error = numeric(n_repeats),
  mean_weighted_sensitivity = numeric(n_repeats),
  mean_weighted_specificity = numeric(n_repeats),
  check.names = FALSE
)

# Repeat process
for (rep in 1:n_repeats) {
  cat("Repeat", rep, "\n")

  # Assign new folds for this repeat
  imp1.2$fold_id <- sample(rep(1:k, length.out = nrow(imp1.2)))

  # Initialise storage
  weighted_aucs <- numeric(k)
  optimal_thresholds <- numeric(k)
  weighted_errors <- numeric(k)
  weighted_sensitivities <- numeric(k)
  weighted_specificities <- numeric(k)

  for (i in 1:k) {
    # Split data
    train_data <- imp1.2 %>% filter(fold_id != i)
    test_data <- imp1.2 %>% filter(fold_id == i)

    # Survey design for training
    train_design <- svydesign(
      id = ~PSU,
      strata = ~Strata,
      weights = ~SurvWeight,
      nest = TRUE,
      data = train_data
    )
  }
}

```

```

# Fit model
model <- svyglm(is_diabetic ~ Almi + Age_yrs + Gender + Race + Bfp_perc + WaistCircum_cm,
               design = train_design,
               family = quasibinomial)

# Get predictions for both train and test sets
train_preds <- predict(model, newdata = train_data, type = "response")
test_preds  <- predict(model, newdata = test_data, type = "response")

# Weighted ROC and AUC (from training data)
roc_obj <- WeightedROC(guess = train_preds,
                      label = train_data$is_diabetic,
                      weight = train_data$SurvWeight)

weighted_aucs[i] <- WeightedAUC(roc_obj)

# Optimal threshold from Youden's J (from training data)
roc_obj$YoudenJ <- roc_obj$TPR - roc_obj$FPR
opt_idx <- which.max(roc_obj$YoudenJ)
optimal_threshold <- roc_obj$threshold[opt_idx]
optimal_thresholds[i] <- optimal_threshold

# Apply optimal threshold to the test data to predict classes
predicted_classes <- ifelse(test_preds >= optimal_threshold, 1, 0)
test_data$predicted_class <- predicted_classes

# Create an error column
test_data$error <- ifelse(test_data$predicted_class != test_data$is_diabetic, 1, 0)

# Now, define the survey design for the test set
test_design <- svydesign(
  id = ~PSU,
  strata = ~Strata,
  weights = ~SurvWeight,
  nest = TRUE,
  data = test_data
)

# Calculate weighted error
weighted_errors[i] <- svymean(~error, design = test_design)[1]

# Calculate Weighted Sensitivity & Specificity
conf_matrix <- svytable(~predicted_class + is_diabetic, design = test_design)

# Extract the four components
TN <- conf_matrix[1, 1]
FN <- conf_matrix[1, 2]
FP <- conf_matrix[2, 1]
TP <- conf_matrix[2, 2]

# Calculate sensitivity and specificity
weighted_sensitivities[i] <- TP / (TP + FN)
weighted_specificities[i] <- TN / (TN + FP)

# Print statement
cat(sprintf(" Fold %d | AUC: %.3f | Thresh: %.3f | Sens: %.3f | Spec: %.3f | Err: %.3f\n",
            i, weighted_aucs[i], optimal_thresholds[i],
            weighted_sensitivities[i], weighted_specificities[i], weighted_errors[i]))
}

# Store means from this repeat
repeat_summary[rep, ] <- c(
  `repeat` = rep,
  mean_weighted_auc = mean(weighted_aucs),

```

```

    mean_optimal_threshold = mean(optimal_thresholds),
    mean_weighted_error = mean(weighted_errors),
    mean_weighted_sensitivity = mean(weighted_sensitivities, na.rm = TRUE), # na.rm is a safeguard
    mean_weighted_specificity = mean(weighted_specificities, na.rm = TRUE)
  )

  # Summary print statement
  cat(sprintf("Repeat %d Summary | Mean AUC: %.3f | Mean Sens: %.3f | Mean Spec: %.3f\n\n",
    rep, repeat_summary$mean_weighted_auc[rep],
    repeat_summary$mean_weighted_sensitivity[rep],
    repeat_summary$mean_weighted_specificity[rep]))
}

# Making table of results

# Long-format data frame for the table that calculates the final average across
# all repeats for each metric.
cv_summary_long <- data.frame(
  Metric = c(
    "AUC",
    "Optimal Threshold",
    "Sensitivity",
    "Specificity",
    "Accuracy"
  ),
  Value = c(
    mean(repeat_summary$mean_weighted_auc),
    mean(repeat_summary$mean_optimal_threshold),
    mean(repeat_summary$mean_weighted_sensitivity),
    mean(repeat_summary$mean_weighted_specificity),
    (1 - mean(repeat_summary$mean_weighted_error)) # Convert error to accuracy
  )
)

# Identify which metrics should be formatted as percentages
percent_metrics <- c("Sensitivity", "Specificity", "Accuracy")

# Create a new, formatted data frame
cv_results_formatted_df <- cv_summary_long %>%
  mutate(
    Value = case_when(
      # For percentage metrics, multiply by 100 and add a '%' sign
      Metric %in% percent_metrics ~ sprintf("%.2f%%", Value * 100),
      # For all others, format to 3 decimal places
      TRUE ~ sprintf("%.3f", Value)
    )
  )

# Build table
cv_summary_tbl_kbl <- kbl(
  cv_results_formatted_df,
  format = "latex",
  booktabs = TRUE
) %>%
  kable_styling(
    font_size = 15,
    full_width = FALSE
  )

# ----- Visualising ROC Curve ----- #

# This helps to convey what maximising Youden's J statistic is doing

# The following code plots the ROC curve from the final loop in the cross validation,

```

```

# with the optimal threshold highlighted whose coordinates were found by maximising
# Youden's J statistic.

# Find the exact coordinates of the optimal point
optimal_point_coords <- roc_obj[which.max(roc_obj$YoudenJ), ]

# Plotting
roc_plot <- ggplot(roc_obj, aes(x = FPR, y = TPR)) +

  # 1. Draw a line connecting all the points
  geom_line(color = "#1f77b4", linewidth = 1) +

  # 2. Add the diagonal reference line
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray") +

  # 3. Highlight single optimal point in orange
  geom_point(data = optimal_point_coords,
             aes(x = FPR, y = TPR),
             color = "#ff7f0e",
             size = 5,
             shape = 18) +

  # Add a label for the optimal point
  annotate("text", x = optimal_point_coords$FPR, y = optimal_point_coords$TPR - 0.05,
          label = "Optimal Threshold",
          color = "#ff7f0e") +

  # Annotate the plot with this fold's AUC value
  annotate("text", x = 0.75, y = 0.25,
          label = paste("AUC =", round(weighted_aucs[k], 3)),
          size = 5) +

  # Add labels and a title
  labs(
    x = "False Positive Rate (1 - Specificity)",
    y = "True Positive Rate (Sensitivity)",
    title = "Progression Model 4 ROC Curve"
  ) +
  theme_dissertation()

# Display the plot
roc_plot

```

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